

Highlights

Tangent-space features for cross-session EEG movement-intent decoding, and the protocol condition that bounds them

Esm E Moula Chowdhury Abha, Md Wahiduzzaman Suva, Muhammad Hasibur Rashid Chayon

- A tangent-space branch exposes channel covariance a log-power stem leaves implicit.
- Accuracy rises on four of five cohorts, 51 participants, three seeds each.
- Highest mean accuracy against eight published decoders run untuned.
- Its label-free reference fails where a class block outlasts the estimator memory.
- Block length against estimator memory, read from the protocol, separates all five.

Tangent-space features for cross-session EEG movement-intent decoding, and the protocol condition that bounds them

Esm E Moula Chowdhury Abha^a, Md Wahiduzzaman Suva^{a,*} and Muhammad Hasibur Rashid Chayon^a

^aDepartment of Computer Science, American International University-Bangladesh, Dhaka, Bangladesh

ARTICLE INFO

Keywords:

brain-computer interface
electroencephalography
test-time adaptation
covariate shift
Riemannian tangent space
calibration
lower-limb exoskeleton

ABSTRACT

Compact EEG decoders summarise each window as the log band power of learned spatial filters, which carry the covariance between channels only implicitly. We add a tangent-space branch beside that stem, mapping each window's shrunk covariance to the tangent space at a reference the network maintains from its input alone. It needs no labels and keeps adapting at test time. Across five cohorts and 51 participants, three seeds each, the branch raises balanced accuracy on four. It adds 0.064 on a 20-participant motor-execution cohort held out from development ($p = 8.8 \times 10^{-5}$, higher in 20 of 20), 0.040 on BCI Competition IV-2a ($p = 0.020$), 0.021 on a second exoskeleton cohort ($p = 0.047$) and 0.045 on a lower-limb exoskeleton cohort ($p = 0.22$); 40 of the 43 participants improve. On those four it has the highest mean accuracy among eight published decoders trained untuned in the same pipeline. A classical tangent-space classifier with no network falls 0.116 below it on the longitudinal exoskeleton cohort and matches it on the held-out cohort. On the fifth cohort the branch costs 0.175, and that failure is predictable before training: the reference is a running mean, safe only while its nominal memory $\mu = N/m$ outlasts a class block. Blocks last 214 windows there against a memory of 160, and 1 to 18 where the branch helps. Moving the memory across the block length switches the effect off and on: +0.045 at 160 windows, +0.048 at 40 and -0.076 at 8. A pre-specified prescription for the rate itself was falsified.

1. Introduction

Spinal cord injury leaves roughly 250,000 to 500,000 people per year with impaired or absent lower-limb function, and powered exoskeletons driven by non-invasive brain signals are one route to restoring overground mobility [1–3]. Decoding the wearer's intent to walk or to stop from scalp EEG is the control problem at the centre of such systems [4, 5]. It is not well described by per-epoch classification accuracy alone, for three reasons that shape the design presented here.

Longitudinal drift The decoder is fitted once and must keep working weeks later. In the primary cohort used here this is literal: the model is fitted on sessions 1–3 and evaluated on sessions 4–9, recorded across weeks. Electrode montage, impedance and skin contact all change between sessions, and the resulting shift in second-order statistics is the dominant source of non-stationarity in EEG decoding [6–8]. Session drift degrades fixed spatial filters and is the reason most BCI systems require a per-session calibration block [9].

Asymmetric cost A false Walk command moves a person's legs. Balanced accuracy scores that identically to a missed Walk and cannot distinguish one sustained false activation from fifty scattered false windows. The quantity a wearer

experiences is spurious activations per minute of standing, which we report alongside accuracy throughout.

What a log-power stem cannot see Compact EEG decoders reduce each window to log power per channel after spatial filtering [10, 11]. That summary is the diagonal of the covariance of the filtered signal. A learned spatial filter $\mathbf{w}$ has output power $\mathbf{w}^\top \mathbf{C} \mathbf{w}$, so the stem is not blind to off-diagonal structure; it represents that structure implicitly, through whichever projections the filters happen to learn, and never holds the full covariance. Covariance structure is the basis of the Riemannian methods that remain competitive on movement-intent tasks [12, 13]. Whether a convolutional decoder is losing usable information at that step is an empirical question, and it is the one this paper answers.

Existing responses to drift are mostly applied offline in preprocessing. Euclidean Alignment whitens each recording by the inverse square root of its mean covariance [14]; Riemannian Procrustes analysis and parallel transport generalise this to the manifold of symmetric positive-definite (SPD) matrices [15–17]. All of these estimate the alignment statistic over a complete recording, which means the target recording must exist before the decoder can run on it. Test-time adaptation removes that requirement in computer vision by updating normalisation statistics or parameters online [18–20], and recent work shows that such methods degrade when the test stream is temporally correlated instead of i.i.d. [21–23]. EEG data acquired under a blocked experimental protocol is temporally correlated by construction, which is the connection this paper makes precise.

Our contributions are:

*Corresponding author

✉ 26-94089-2@student.aiub.edu (E.E.M.C. Abha);

26-94088-2@student.aiub.edu (M.W. Suva); chayon@aiub.edu (M.H.R. Chayon)

ORCID(s):

1. Add a tangent-space branch beside the log-power stem: each window's covariance is shrunk, mapped to the tangent space at a reference estimated from the input alone, and projected into the classifier, with no labels used at inference (Section 4.2).
2. Measure it on five cohorts and 51 participants with three seeds each. Accuracy rises on four, including +0.064 in 20 of 20 participants of a cohort held out from architecture and pipeline development (Section 6.5).
3. Compare against eight published decoders trained in one pipeline with shared splits, seeds and metric code, and report the cohort where most of them fail to train rather than letting the resulting margins stand (Section 6.8).
4. Identify the branch's failure mode and relate it to a quantity read from the protocol. The reference is a running mean over incoming windows, and it degrades when a class block outlasts its memory; on the cohort where a block outlasts that memory the branch costs 0.175 and falls near chance (Condition 1, Section 6.3).
5. Perturb the estimator memory directly at fixed block length. Shrinking it below the block length reverses the gain on cohort A, and enlarging it on cohort B recovers 74 % of the loss without restoring parity, both in the direction the account predicts (Section 6.5).
6. Report as falsified a stronger version of the same condition, one that sets the adaptation rate from the block length: at the memory it prescribes the failing cohort reaches 0.650 against 0.744 for the stem alone (Section 6.5).

The branch costs parameters. It adds 238,028 to a 19,505-parameter stem path, plus 32,896 for the fusion it requires, so the reported model is no longer compact and we do not claim parameter efficiency; Section 6.9 reports the counts beside the accuracies so the trade is visible.

2. Related work

2.1. Compact convolutional decoders for EEG

ShallowFBCSPNet and Deep4Net established that a square $\rightarrow$ pool $\rightarrow$ log pipeline over learned spatial filters recovers the band-power features that filter-bank common spatial patterns computes explicitly [10]. EEGNet reduced the parameter count by an order of magnitude using depthwise-separable convolutions [11], and EEGNeX refined the same family [24]. These architectures work well within a session and are the standard baselines in the field [25, 26]. Their limitation for our setting is that the learned spatial filters are fixed at training time, so a montage or impedance change between sessions is not compensated at all. We keep such a stem unchanged and add a second path beside it that reads each window's covariance relative to a reference estimated on the current recording (Section 4.2).

2.2. Attention-based decoders

EEG Conformer pairs a ShallowFBCSP-style stem with a transformer encoder [27], and EEG-TCNet applies temporal convolutions to the same problem [28]. The stem-plus-transformer combination is already published: an independent implementation of it in our pipeline reproduced EEG Conformer's score without improving on it (0.848 against 0.843), and we therefore claim no novelty for that pairing. What these models share with the convolutional family is a fixed front end, which is the property the present work targets.

2.3. Alignment and transfer in BCI

Riemannian methods treat the spatial covariance as a point on the SPD manifold and classify with geodesic distances [12, 13, 29]. For transfer, Euclidean Alignment whitens each subject or session by $\mathbf{M}^{-1/2}$ where $\mathbf{M}$ is its mean covariance [14]; Riemannian Procrustes analysis adds rotation and scaling [15], and parallel transport moves covariances between tangent spaces [16, 17]. These methods are effective and label-free, and reviews report them as the strongest general-purpose transfer tools for EEG [30, 31]. Their limitation is operational. The alignment statistic is computed over a whole recording, so the recording must be complete before decoding starts, and within-recording drift is not tracked. Online variants remove that requirement: EDAPT updates a reference covariance by an exponential moving average after each trial and whitens incoming trials by its inverse square root [32], and the running reference used here is of the same kind. What this paper adds is not the online estimator but a second-order branch built on it, and an account of when such an estimator fails, which depends on the temporal structure of the test stream (Section 6.3).

2.4. Riemannian and tangent-space representations

Spatial covariances are symmetric positive-definite matrices and do not form a vector space, so the methods built on them operate on the SPD manifold. The standard recipe computes a covariance per window, maps it to the tangent space at a reference point, and classifies the resulting vector with a linear model [12, 13, 29]. The reference is usually the Riemannian or arithmetic mean of the training covariances, and the choice matters because the map is only a faithful local approximation near that point. These representations remain competitive with convolutional decoders on movement-intent tasks and are the strongest general-purpose transfer tools reported in reviews [30, 31]. SPD networks learn the manifold operations instead of fixing them [33].

Two differences separate the branch used here from that line. The tangent map is a second input to a convolutional decoder rather than the representation the decoder is built on, so first-order band power and second-order structure are read together instead of one replacing the other; Section 6.5 reports that their fusion exceeds the stem alone; a decoder built on the branch alone was not trained. And the reference point is estimated from the incoming windows at inference

rather than fixed after fitting, which is what makes the representation track a changing session and is also, as Section 6.3 shows, what makes it fail under a blocked protocol.

2.5. Test-time adaptation

Updating normalisation statistics on the test distribution improves accuracy under image corruption in vision [19, 20], and Tent extends this to updating affine parameters by entropy minimisation [18]. Test-time training adds a self-supervised objective at inference [34]. A recurring finding in this literature is that these methods assume the test stream is i.i.d., and degrade when it is not: NOTE addresses temporally correlated streams by maintaining a class-balanced reservoir [21], SAR stabilises adaptation under distribution shift with small or imbalanced batches [22], and CoTTA and LAME address error accumulation in continual adaptation [23, 35]. Our contribution relative to this line is quantitative: we show that in a blocked EEG protocol the relevant comparison is between the estimator's nominal memory and the duration of a single-class block, and that this comparison predicted a new cohort in advance. The constraint should also have an upper side, because the same estimator is what tracks the drift the layer exists to remove; that side is not measured here (Section 8).

2.6. EEG correlates of gait and exoskeleton control

Walking modulates cortical activity in a gait-phase-locked manner [36, 37], and treadmill walking can be decoded from scalp EEG [5, 38]. Robot-assisted gait modulates midline sensorimotor rhythms [39]. The intent signal our decoder targets is event-related desynchronisation in the mu and beta bands, a power decrease over sensorimotor cortex preceding and accompanying movement [40, 41]. This motivates a representation in which log-power ratios are preserved (Section 4.1), because ERD is defined as a relative power change.

3. Problem formulation

Let $\mathbf{X} \in \mathbb{R}^{C \times T}$ be a window of C channels and T samples with label $y \in \{\text{Stop}, \text{Walk}\}$. A recording session s supplies windows drawn from $p_s(\mathbf{X}, y)$. We assume the conditional $p(y | \mathbf{X})$ is stable across sessions but the marginal $p_s(\mathbf{X})$ is not, which is the covariate-shift assumption standard in this literature [6, 42]. Concretely we take the shift to act primarily on the spatial covariance $\Sigma_s = \mathbb{E}_{p_s}[\mathbf{X}\mathbf{X}^\top]/T$.

The decoder is fitted on sessions S_{fit} and evaluated on disjoint later sessions S_{test} , with no labels available from S_{test} at any point. This is the constraint that rules out per-session recalibration and motivates a label-free correction.

We quantify session drift by the affine-invariant Riemannian distance between SPD matrices,

$$\delta(\mathbf{A}, \mathbf{B}) = \left\| \log(\mathbf{A}^{-1/2} \mathbf{B} \mathbf{A}^{-1/2}) \right\|_F, \quad (1)$$

in preference to the Frobenius distance, because covariances do not form a vector space and the Frobenius distance is

dominated by overall scale, which would flatter any whitening operation [12, 13].

Two timescales govern the analysis in Section 6.3. Let τ_{blk} be the median duration, in windows, of a contiguous single-class block in the recording protocol, and let τ_{drift} be the timescale over which Σ_s changes appreciably. Both are properties of the experimental design and the recording hardware, independent of the model.

4. Method

The reported decoder has two components. A multi-scale log-power stem represents each window by the band power of learned spatial filters, which carries the covariance between channels only implicitly, and a tangent-space branch represents the full regularised covariance explicitly. Their outputs are concatenated and passed to one linear classifier. The stem path carries 19,505 parameters and the branch 238,028, with a further 32,896 in the fusion layer the branch requires, so the branch is the larger part of the model by an order of magnitude and Section 6.9 reports what that buys.

Three further components were built and tested during development and none is retained. They are not evaluated in this paper, and the ablations of Section 6.1 concern only the two paths above.

4.1. Multi-scale power stem

Three parallel temporal resolutions (64, 128 and 256 ms), each with its own depthwise spatial filters, are followed by square $\rightarrow$ pool $\rightarrow$ log, after Schirrmester et al. [10]. Amplitude normalisation precedes the squaring so that log-power ratios survive: scaling a window by s shifts its log-power by $2 \log s$, a constant that a later affine layer absorbs, whereas normalising after the square destroys the ratio in which ERD is defined [40].

Each temporal resolution is a convolution with 16 kernels of 6, 13 or 26 samples at 100 Hz, without bias, followed by batch normalisation, a depthwise convolution across all channels with two spatial filters per kernel, and batch normalisation again. Squaring, non-overlapping average pooling over 200 ms frames and a logarithm with a floor of 10^{-6} give 32 log-power features per resolution and 96 per frame. After dropout of 0.3 and layer normalisation, each frame is projected to 128 dimensions with a GELU, and the frames of a window are pooled by a learned softmax attention over time into one 128-dimensional embedding.

This ordering is load-bearing, and the evidence is a dose-response across representations that differ in how much marginal power they carry (Table 4). Removing per-window amplitude normalisation is worth +0.134 and +0.174 on band-power features, which are pure marginal power; −0.016 and +0.029 on tangent-space features, which carry power only through the covariance diagonal; and −0.004 and +0.000 on a correlation-only representation, which is scale-free by construction and therefore cannot respond. A representation that carries no marginal power shows no effect, which is what identifies the mechanism instead of merely demonstrating it.

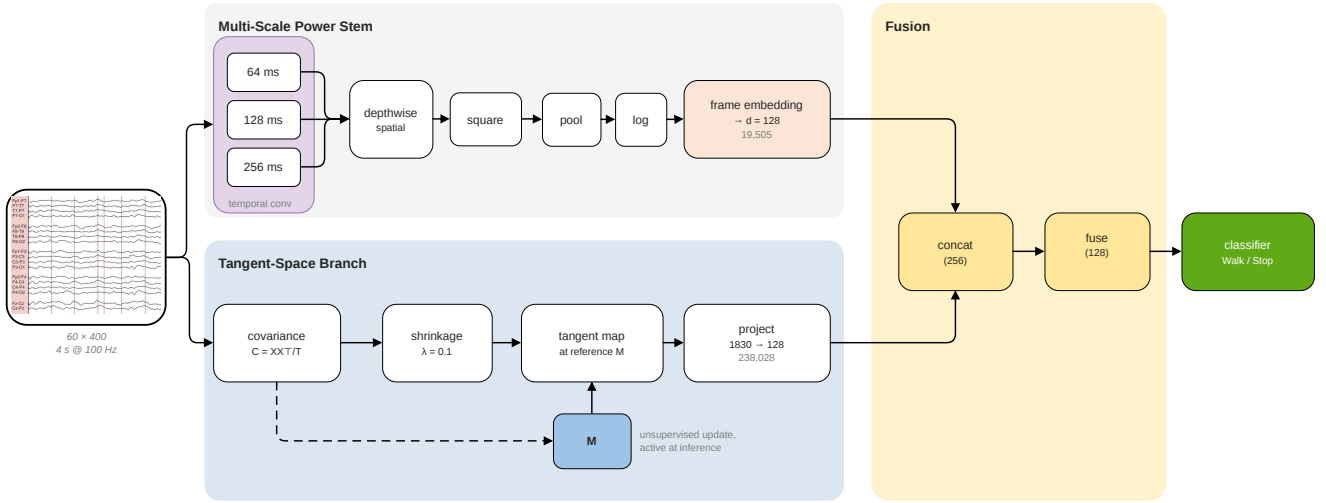


Figure 1: The reported decoder. A window enters both paths: the stem gives the log band power of learned spatial filters, and the branch maps the window’s covariance to the tangent space at a running reference and projects the result. The two are concatenated into one linear classifier. The dashed arrow is the reference update, which uses no labels and stays active at inference.

These are participant-paired comparisons, so they can be tested as such. Wilcoxon signed-rank tests over participants, Holm-corrected across the three representations within each cohort, give band power +0.174 (better in 7 of 8, $p_{\text{Holm}} = 0.047$) on cohort B and +0.134 (6 of 7, $p_{\text{Holm}} = 0.094$) on cohort A; tangent space +0.029 (5 of 8, $p_{\text{Holm}} = 0.500$) and −0.016 (2 of 7, $p_{\text{Holm}} = 0.750$); and the correlation-only control +0.000 (3 of 8, $p_{\text{Holm}} = 1.000$) and −0.004 (3 of 7, $p_{\text{Holm}} = 0.750$). **The effect reaches significance on the larger cohort and is null in the scale-free control, as the mechanism predicts;** cohort A is consistent in sign but underpowered at $n = 7$, where the smallest attainable Holm-corrected p is 0.047.

A separate sweep over where normalisation is applied in the network confirms the ordering at the architecture level: on cohort A with 4 s windows, per-window normalisation scores 0.849, global per-recording normalisation 0.878, and no normalisation 0.864, so the global setting used here is the best of the three and per-window normalisation is the worst.

Table 5 repeats the normalisation comparison across six published decoders, in an earlier training pipeline. Per-window normalisation is the best choice for two of them and the worst for two, and most differences are below the 0.02 run-to-run variation, so this comparison neither supports nor contradicts the representation-level dose–response in Table 4. Table 4 also reports the control that retired an amplitude side-channel: supplying a per-window amplitude feature and supplying the same feature with its window assignment shuffled differ by less than 0.02 for four of five models.

4.2. Tangent-space branch

The stem of Section 4.1 sees the covariance between channels only through its learned spatial filters: a filter $\mathbf{w}$ has output power $\mathbf{w}^T C \mathbf{w}$, which depends on every entry of the covariance C , but only along the directions the filters have

learned. The branch exposes the full covariance explicitly. For a window X , mean-centred over time, we form $C = XX^T/T$, normalise its trace and shrink it toward the identity,

$$\tilde{C} = (1 - \lambda) \frac{C}{\text{tr}(C)/n_{\text{ch}}} + \lambda I, \quad \lambda = 0.1, \quad (2)$$

and map it to the tangent space at a reference M ,

$$\phi(X) = \text{vec}(\log(M^{-1/2} \tilde{C} M^{-1/2})), \quad (3)$$

with off-diagonal entries scaled by $\sqrt{2}$ so that the vector’s Euclidean norm equals the matrix Frobenius norm. Both matrix functions are computed by symmetric eigendecomposition in double precision, with eigenvalues floored at 10^{-6} before the inverse square root and the logarithm. The vector has $n_{\text{ch}}(n_{\text{ch}} + 1)/2$ entries, 1830 for 60 channels, and passes through layer normalisation and a linear projection to 128 dimensions with a GELU and dropout of 0.3. It is concatenated with the stem’s 128-dimensional embedding, and a linear fusion layer maps the pair back to 128 dimensions before the classifier, which leaves the classifier unchanged in shape so the branch ablates as one module.

The reference is a running mean over the covariances of incoming windows, each trace-normalised and shrunk by (2) before being accumulated, so the estimate is an average of shapes rather than an average weighted by power:

$$\mathbf{M} \leftarrow (1 - m) \mathbf{M} + \frac{m}{N} \sum_{n=1}^N \tilde{C}_n, \quad (4)$$

where $\tilde{C}_n$ is (2) applied to the n -th window of an update, N the number of windows in an update and m the momentum. We use $\mu = N/m$ windows as the estimator’s nominal memory. It is a characteristic scale rather than a buffer length: the weight of past updates decays geometrically, and

at $m = 0.2$ the e-folding time is $-1/\ln(1-m) \approx 4.5$ updates, or 143 windows at $N = 32$, against the nominal 160. Under either definition cohort B's 214-window block exceeds the memory at the default rate and every other cohort's median block falls well below it, so the comparisons that follow do not depend on the choice. Without the trace normalisation the estimate is power-weighted, and on a cohort whose majority class carries $1.52\times$ the power of the minority class a power-weighted estimate is the majority class rather than an average over both.

The reference M is a running mean over the covariances of incoming windows, updated by (4) with momentum m and no labels, and it keeps updating at inference. This is what lets the representation follow a session as it changes, and it is also the component that fails: because M is an average over whatever is currently streaming, a protocol that presents one class for longer than the estimator remembers will drive M onto that class. Condition 1 states when that should not happen, and Section 6.5 measures it on five cohorts.

Update order at evaluation The reference is initialised from the first training batch and is not reset afterwards: every forward pass updates it, in fitting, in the held-out validation passes and in calibration, so it enters each participant's test stream carrying that history. Test windows are presented in recording order, in consecutive batches of $N = 32$, and within a forward pass the reference is updated with the batch before it is applied to that same batch. Each test window's reference therefore includes the covariances of up to 31 later windows of its batch, about 15 s at a 0.5 s stride, so in this respect too the evaluation is offline. A causal implementation would compute the representation with the reference from before the batch and update it afterwards. That variant was not run, and whether the reported effects depend on the look-ahead is not measured here.

The tangent map carries no gradient; only the projection is learned. Backward through an eigendecomposition contains $1/(\lambda_i - \lambda_j)$ terms that diverge on near-degenerate spectra, and covariances estimated from 60 channels and 400 samples are routinely near-degenerate. Nothing measured is lost by detaching it: the probe that motivated the branch was itself a linear map on fixed tangent features, and that is what is trained here, jointly with the network rather than after it. The branch carries 238,028 parameters and the fusion layer it requires a further 32,896, against 19,505 in the stem path and 514 in the shared classifier; the reported model is 290,943 in total, of which the branch and its fusion are 93 %. Counts quoted for a path exclude the classifier; counts quoted for a model include it, so the stem on its own is 20,019.

5. Experimental setup

5.1. Data

Cohort A (exoskeleton, OpenNeuro ds007788). Seven participants, nine sessions each, recorded across weeks while walking in a powered lower-limb exoskeleton [4]. Sixty-channel active EEG at 100 Hz; walk/stop labels are taken from the exoskeleton controller state. Windows of

4 s at 0.5 s stride give approximately 5031 fitting windows per participant. The model is fitted on sessions 1–3 and evaluated on sessions 4–9.

Cohort B (treadmill MoBI). Eight participants, three trials each, 64-channel EEG at 100 Hz, walk/stand from treadmill phase [38]. Fitted on trial 1 and evaluated on trials 2–3, with 2 s windows. The cohort is from an independent laboratory with different sensors, a different task framing, and reversed class balance: Stop is the minority class at approximately 12 % here and the majority class in cohort A.

Cohort C (motor execution, PhysioNet EEGMIDB). Twenty participants, S001–S020, from the PhysioNet motor movement and imagery database [43, 44], recorded with 64-channel EEG at 160 Hz and resampled to 100 Hz. Only the motor-execution runs are used, and the task is rest against movement of the hands or feet. The model is fitted on runs 3, 5, 7 and 9 and evaluated on runs 11 and 13, with 2 s windows at 0.5 s stride, each lying wholly inside one annotated period. The protocol alternates the two classes about every 4 s, a median of 5 windows per class block. Its analysis plan was committed to version control before any model was trained on it.

Cohort D (BCI Competition IV-2a, BNCI2014-001). [45] Nine participants recorded in two sessions on different days, 22-channel EEG at 250 Hz resampled to 100 Hz, cued motor imagery of the left against the right hand. The model is fitted on the first session and evaluated on the second, with 4 s windows covering the imagery interval, 288 trials per participant and balanced classes. Trial order is randomised, so a single-class block is one trial: this is the cohort where a covariance update spans the largest number of blocks. It contributes about 100 fitting windows per participant, the fewest of the five, which Section 6.8 shows breaks most of the published decoders under shared hyperparameters.

Cohort E (lower-limb exoskeleton, DECODED). Seven participants wearing a powered lower-limb exoskeleton, 27-channel EEG at 200 Hz resampled to 100 Hz, from the DECODED subproject of EUROBENCH [46]. Each run presents 15 s standing relaxed with the device stopped, 24 s walking under kinesthetic motor imagery, 22 s walking under a regressive-counting distractor and 14 s standing relaxed. We decode standing relaxed against walking under motor imagery, the same contrast as cohort A on a second device, and drop the counting condition rather than folding it into either class, since it is a distractor rather than a movement-intent state. Windows are 4 s at 0.5 s stride and never cross a condition boundary. Only one participant recorded on two dates, so unlike cohorts A, C and D the split is temporal across runs, the first eight fitting and the rest testing. This cohort therefore matches best on task and tests least drift, and Section 8 says so.

The protocols differ sharply in temporal structure, which Section 6.3 shows to be decisive (Table 1).

τ_{blk} is a median, and Table 1 also shows that cohort A contains single blocks of 486 windows, longer than the 160-window memory the default rate gives. The two are consistent because what the reference tracks is typical exposure

Table 1

Contiguous single-class block structure, computed from the label sequence alone by `tools/tau_blk_rule.py`. Blocks are counted within a recording, never across one, and pooled over participants; τ_{blk} is the median block duration in windows. It is a property of the protocol and not of the participant: every participant of every cohort returns the same median.

Cohort	Blocks	τ_{blk}	Longest
A (exoskeleton)	789	18	486
B (treadmill)	66	214	2211
C (motor execution)	3600	5	5
D (BCI IV-2a)	734	1	6
E (exoskeleton, DECODED)	192	13	33

and not the worst block: on cohort A, 2 of 789 blocks exceed the memory and they hold 4.8 % of the windows, with the 95th percentile block at 112 windows. A block that outlasts the memory pulls the reference toward its class while it streams, and the reference returns once the protocol alternates again, so on that cohort the corruption is a rare excursion. On cohort B the median itself exceeds the memory, so the same corruption is the normal state rather than an excursion, which is the distinction the condition turns on. The median is therefore the operative statistic, and a protocol whose median sits below the memory but whose long blocks dominate the recording would not be covered by the evidence here.

5.2. Preprocessing and training

Signals are band-pass filtered to 8–30 Hz and resampled to 100 Hz. Cohort A uses a zero-phase FIR filter in MNE-Python [47], after which the EOG channels, filtered to the same band, are regressed linearly out of the EEG. Cohorts B and C use a zero-phase fourth-order Butterworth filter and have no EOG step. Zero-phase filtering is acausal: each filtered sample depends on samples that follow it. The amplitude normalisation takes its statistics from the fitting split alone, but the band-pass is acausal, and so is the order in which the running reference is updated within a batch (Section 4.2), so the evaluation reported here is an offline one. A deployed version would substitute a causal band-pass, which the results below do not measure. Each recording, or each run where a cohort is split by runs, is filtered on its own, so no filter spans the boundary between fitting and test data. No other artefact rejection, ICA decomposition or channel interpolation is applied, so that the running reference is evaluated on the shift it is meant to track. The artefact control of Section 6.2 is the one exception and is reported separately. The band also excludes the gait cycle, near 1–2 Hz, and most electromyographic power above 30 Hz. Amplitude normalisation is applied globally, before the squaring stage, for the reason given in Section 4.1. The per-channel mean and standard deviation are computed on the fitting split alone and then applied unchanged to the calibration and test splits, so no statistic is taken from the recordings the model is evaluated on.

Table 2

Training configuration, identical across all arms, all cohorts and the published decoders of Table 3, except where the cohort definition differs.

Setting	Value
Band-pass / resample	8–30 Hz, 100 Hz (zero-phase)
Artefact correction	EOG regression (A); none (B–E)
Amplitude normalisation	global, fitting-split statistics
Window / stride	4 s / 0.5 s (A, E); 2 s (B); 2 s / 0.5 s (C); 4 s, one per trial (D)
Optimiser	AdamW [48], lr 3×10^{-4} , weight decay 10^{-2} , one-cycle schedule (30 % warm-up, cosine annealing)
Batch size N	32
Epochs	at most 30; early stopping after 12 without improvement
Gradient clipping	norm 1.0
Loss	cross-entropy, inverse-frequency weighted
Adaptation momentum m	0.2, nominal memory $N/m = 160$ windows (swept in §6.3)
Held-out split	about 30 % of whole segments per class, fitting data only
Seeds	0, 1, 2: held-out split, initialisation and batch order
Model selection	best held-out balanced accuracy
Calibration	temperature scaling fitted on the held-out split

Table 2 lists the training configuration. The adaptation momentum is $m = 0.2$ unless stated otherwise; Section 6.3 treats m as the object of study. Model selection uses balanced accuracy on a held-out split of the fitting data, evaluated at the best epoch by that criterion, which is conventional but upward-biased on small folds (Section 8). The held-out split takes whole segments, contiguous stretches of one recording and one class, choosing about 30 % of each class's segments at random, so overlapping windows from one segment never fall on both sides. On cohort A it is drawn from the training-task recordings of the fitting sessions, and the closed-loop trial recordings are used for fitting only. A data-split seed (0, 1 or 2) sets which segments are held out, the weight initialisation and the batch order; class weights and normalisation statistics come from the fitting data alone.

Published decoders Eight published decoders from the *braindecode* library [49] were trained in the same pipeline as the decoder: ShallowFBCSPNet and Deep4Net [10], EEGNet [11], EEGNetX [24], EEG Conformer [27], TSception [50], FBLightConvNet [51] and EEG-TCNet [28]. TSception was designed for emotion recognition and is used here as a general EEG backbone, not as an established movement-intent decoder. Each model's classification layer is replaced by the same linear projection and classifier, and each is trained with the preprocessing, optimiser, schedule, early stopping, model selection and temperature scaling of Table 2, so that the models differ only in the backbone. No

per-model tuning was performed. Parameter counts include the shared classifier and are instantiated at cohort A's input shape.

5.3. Outcome measures

Balanced accuracy for comparability with the literature; expected calibration error (ECE) with 10 equal-width bins [52], because a device acting on a confidence value needs that value to mean something; and spurious activations per minute of standing.

Both onset quantities are computed from the decided label sequence, with no threshold, hysteresis or merging: the decision is the argmax of the temperature-scaled posterior, and quantities are accumulated within a recording and never across one. A *false onset* is a $0 \rightarrow 1$ transition of the predicted sequence at a window whose true label is the negative class; the rate divides the count by the duration of the negative-class windows, each window contributing its 0.5 s stride. A *missed onset* is a true $0 \rightarrow 1$ transition for which no window is predicted positive between that transition and the end of the positive block it opens; the reported figure is the fraction of true onsets that are missed. Detection latency is not reported: the inference cost of Section 6.7 measures computation, which is not the same quantity, and the model's 4 s windows set a floor on detection that this evaluation does not resolve.

Leakage guard Unconditioned nuisance-decodability is confounded with accuracy. In our pipeline a decoder that uses no movement information whatever measures $R^2 = +0.863$ against the movement signal, because both the decoder output and the nuisance are driven by the class label. Movement leakage is therefore measured conditional on the class label throughout, by centring the nuisance features within class before regression.

We calibrate the probe by injecting the recorded movement signal into the EEG, spatially mixed with a fixed per-participant gain matrix, at a known fraction f of signal power. The calibration uses the tangent-space representation. Accuracy rises monotonically with f : 0.763 at $f = 0$, then 0.819, 0.873, 0.892 and 0.913 at $f = 0.001, 0.005, 0.01$ and 0.05. The conditional probe rises with it, from -0.112 at $f = 0$ through $+0.015$ at $f = 0.005$ to $+0.067$ at $f = 0.02$, where accuracy is 0.903. Reaching an accuracy near 0.90 through injected movement therefore moves the probe to about $+0.07$. The reported model reaches 0.909 on cohort A with a conditional probe of -0.074 , and the stem reaches 0.864 with -0.080 , both on the uncontaminated side of the calibration curve. Two limits apply. The calibration was performed on a different representation from the decoder it is compared with, and the conditional probe cannot register artefact information that is constant within each class. The comparison is therefore evidence against movement-driven accuracy. Section 6.2 adds a second, independent test by removing artefact components and retraining.

5.4. Statistical treatment and measurement noise

Accuracy comparisons are reported as the mean over participants with the between-subject standard deviation, and as paired Wilcoxon signed-rank tests over participants wherever per-participant results were retained (Section 8). The five cohorts are independent datasets, each testing the same pre-stated hypothesis that the branch raises accuracy over the stem, so the cohort-level tests are reported as separate replications and are not corrected across cohorts. Comparisons made within one cohort, such as the eight published decoders against the reported model, form a family on the same data and are Holm-corrected. Every $\pm$ in a table is labelled in its caption as across participants or across data-split seeds. Standard deviations across participants are population values, and standard deviations across seeds are sample values.

The quantity that governs whether two models can be ranked is the spread across data-split seeds, and it is not uniform across arms or cohorts (Table 6). Over three seeds with every other setting fixed, the reported model spans 0.0070 to 0.0216 depending on the cohort, and the stem spans 0.0110 to 0.0569; the widest is the stem on cohort A. The stem is the noisier arm on all five cohorts, so the branch does not buy its accuracy with a less stable fit. Differences below 0.02, the order of that run-to-run variation, and differences smaller than the relevant seed spread are reported without a ranking. This is a descriptive rule from three seeds, not a statistical criterion; the conclusions rest on the paired tests across participants. The effects we do interpret are measured over three data-split seeds and tested paired within participant (Tables 3 and 12).

Pipeline separation The proposed decoder and the published decoders of Section 6.9 were trained in one pipeline. Tables 4 and 5 come from earlier pipelines and are used only for comparisons within each table.

6. Results

6.1. The branch against the stem

Table 7 and Fig. 4 give the branch against the stem it is added to, on cohort A, over three data-split seeds. Accuracy rises from 0.864 to 0.909, higher in 6 of 7 participants. At seven participants the paired difference is not significant ($p = 0.22$), and the cohorts that carry the effect are the larger ones in Section 6.5.

The seed column is worth as much as the accuracy column. The stem's grand mean moves by 0.032 across seeds and the branch's by 0.0036, so across the three seeds evaluated here the branch shows substantially lower seed-to-seed variability on this cohort. That matters for every comparison in this paper, because the threshold below which differences are not interpreted was set by seed spread.

Table 3

The reported model, the stem it is built on, and eight published decoders trained in one pipeline with identical preprocessing, optimiser, schedule, early stopping, model selection and classifier head. Balanced accuracy, mean $\pm$ SD across three data-split seeds; cohorts A to E have 7, 8, 20, 9 and 7 participants. Bold marks a configuration of ours that exceeds every published decoder in that column. Parameter counts include the shared classifier at cohort A's input shape. † near chance (< 0.55) for at least five participants on a seed.

Model	Params	A	B	C	D	E
<i>This work</i>						
stem + branch	290 943	0.909 $\pm$ 0.004	0.568 $\pm$ 0.005	0.843 $\pm$ 0.006	0.741 $\pm$ 0.011	0.949 $\pm$ 0.005
stem alone	20 019	0.864 $\pm$ 0.032	0.744 $\pm$ 0.016	0.779 $\pm$ 0.007	0.701 $\pm$ 0.023	0.928 $\pm$ 0.006
<i>Published decoders, same pipeline</i>						
EEGNeX [24]	71 010	0.876 $\pm$ 0.014	0.707 $\pm$ 0.023	0.783 $\pm$ 0.004	0.656 $\pm$ 0.015	0.887 $\pm$ 0.018
EEGConformer [27]	445 442	0.847 $\pm$ 0.008	0.685 $\pm$ 0.018	0.783 $\pm$ 0.014	0.599 $\pm$ 0.077†	0.896 $\pm$ 0.010
FBLightConvNet [51]	23 760	0.866 $\pm$ 0.011	0.687 $\pm$ 0.010	0.778 $\pm$ 0.006	0.674 $\pm$ 0.004	0.891 $\pm$ 0.022
ShallowFBCSPNet [10]	206 962	0.849 $\pm$ 0.020	0.692 $\pm$ 0.027	0.765 $\pm$ 0.004	0.682 $\pm$ 0.013	0.886 $\pm$ 0.008
TSception [50]	182 098	0.852 $\pm$ 0.003	0.734 $\pm$ 0.017	0.723 $\pm$ 0.001	0.579 $\pm$ 0.015†	0.833 $\pm$ 0.011
EEGNet [11]	27 666	0.856 $\pm$ 0.028	0.716 $\pm$ 0.025	0.630 $\pm$ 0.049†	0.526 $\pm$ 0.017†	0.700 $\pm$ 0.054
EEGTCNet [28]	7 062	0.878 $\pm$ 0.009	0.672 $\pm$ 0.024	0.585 $\pm$ 0.037†	0.505 $\pm$ 0.015†	0.695 $\pm$ 0.016
Deep4Net [10]	717 417	0.817 $\pm$ 0.014	0.617 $\pm$ 0.030	0.549 $\pm$ 0.004†	0.525 $\pm$ 0.007†	0.722 $\pm$ 0.035

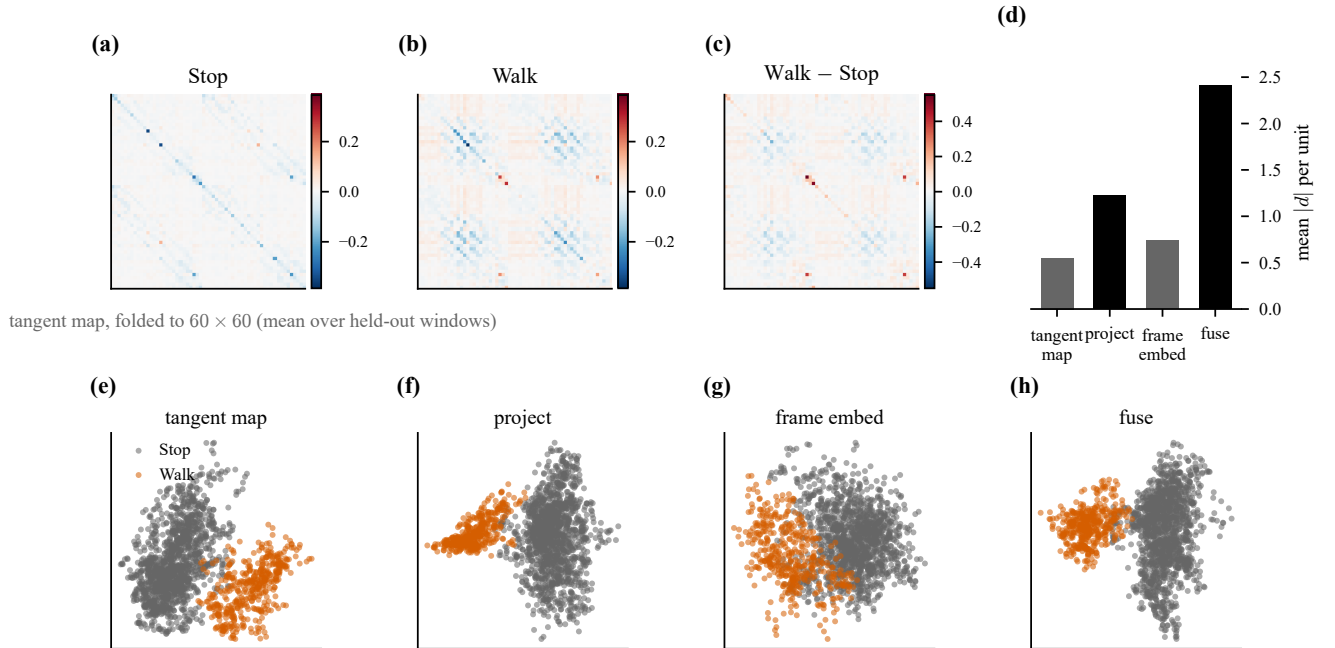


Figure 2: What the model computes, measured rather than drawn. Forward hooks recorded activations at the four points Fig. 1 names, on held-out windows of one cohort A participant. (a–c) The tangent vector folded back into the 60×60 matrix it came from, for Stop, Walk and their difference. (d) Class separation at each tap. (e–h) The same four representations in their first two principal components.

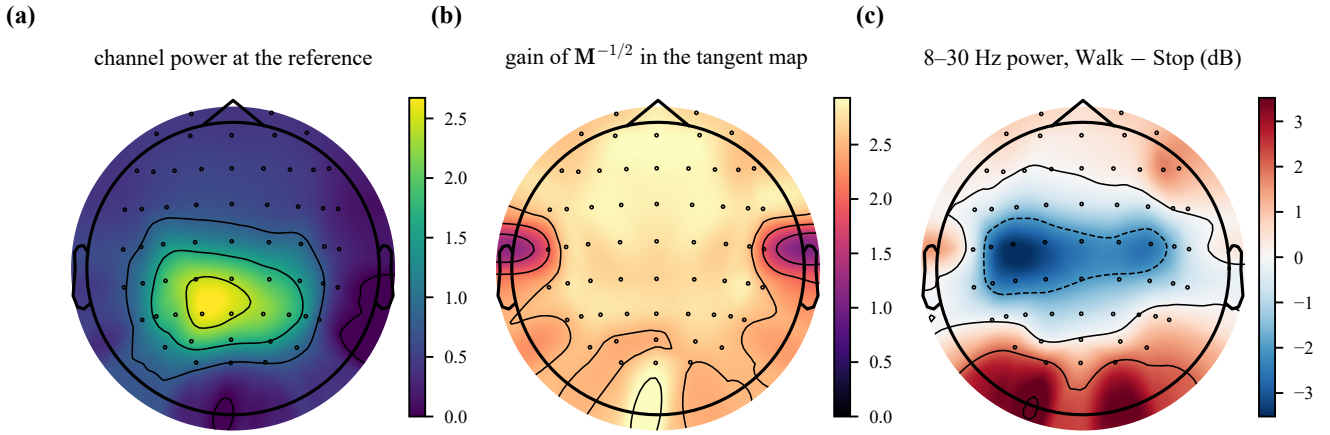


Figure 3: Where the branch's reference acts, and what the classes differ by. One cohort A participant, first three sessions. The reference is built by the branch itself, so these are quantities the reported model holds at inference. (a) Channel power at the reference. (b) Per-channel gain of $M^{-1/2}$, which the tangent map applies. (c) Absolute 8–30 Hz power, Walk minus Stop.

Table 4

Two-condition comparisons. The first block removes per-window amplitude normalisation across representations that differ in how much marginal power they carry. The second supplies a per-window amplitude feature, and the same feature with its window assignment shuffled. Single seed; condition and control accuracies are mean $\pm$ SD across participants.

Comparison	Power content	Condition	Control	Δ
<i>Per-window amplitude normalisation, by how much marginal power the representation carries</i>				
band power, cohort A	full	0.625 ± 0.091	0.760 ± 0.113	+0.134
band power, cohort B	full	0.574 ± 0.084	0.748 ± 0.067	+0.174
tangent space, cohort A	partial	0.762 ± 0.146	0.746 ± 0.138	-0.016
tangent space, cohort B	partial	0.675 ± 0.088	0.704 ± 0.085	+0.029
correlation, cohort A	none	0.738 ± 0.138	0.734 ± 0.134	-0.004
correlation, cohort B	none	0.701 ± 0.058	0.702 ± 0.058	+0.000
<i>Amplitude side-channel against its shuffled control</i>				
ShallowFBCSPNet	-	0.832 ± 0.078	0.816 ± 0.101	-0.017
Deep4Net	-	0.798 ± 0.142	0.809 ± 0.123	+0.011
EEGNetX	-	0.841 ± 0.093	0.841 ± 0.086	-0.000
EEGNet	-	0.828 ± 0.115	0.837 ± 0.111	+0.008
BDTCN	-	0.748 ± 0.136	0.652 ± 0.151	-0.095

Table 5

Where amplitude normalisation is applied, across 6 published decoders on cohort A, single seed, in an earlier training pipeline. All three columns are balanced accuracies, mean $\pm$ SD across participants; best mean per row in bold.

Model	Per-window	Global	None
ShallowFBCSPNet [10]	0.824 $\pm$ 0.098	0.821 $\pm$ 0.087	0.818 $\pm$ 0.096
Deep4Net [10]	0.760 $\pm$ 0.090	0.821 $\pm$ 0.131	0.821 $\pm$ 0.121
EEGNetX [24]	0.834 $\pm$ 0.085	0.832 $\pm$ 0.104	0.835 $\pm$ 0.097
EEGNet [53]	0.833 $\pm$ 0.096	0.791 $\pm$ 0.134	0.795 $\pm$ 0.137
BDTCN [54]	0.820 $\pm$ 0.048	0.822 $\pm$ 0.080	0.820 $\pm$ 0.056
EEGConformer [27]	0.828 $\pm$ 0.066	0.829 $\pm$ 0.079	0.805 $\pm$ 0.074

6.2. Artefact control

Walk and stop differ in movement, so accuracy on this task could arise from muscle or motion artefact. The reported configuration was therefore retrained after independent component analysis, with both arms restricted to the training task so that they differ only in the cleaning step. Each recording was re-referenced to the average, a copy band-passed to 1–45 Hz was decomposed into 30 components with

Table 6

Measurement noise, for the arms this paper reports. Spread is the range of balanced accuracy across three data-split seeds with every other setting fixed.

Arm	n	Spread
<i>Across three data-split seeds, reported arms</i>		
stem + branch, cohort A	3	0.0070
stem + branch, cohort B	3	0.0088
stem + branch, cohort C	3	0.0108
stem + branch, cohort D	3	0.0216
stem + branch, cohort E	3	0.0091
stem, cohort A	3	0.0569
stem, cohort B	3	0.0300
stem, cohort C	3	0.0132
stem, cohort D	3	0.0455
stem, cohort E	3	0.0110
<i>Across data-split seeds, published decoders, same pipeline</i>		
Deep4Net	3	0.0265
EEGConformer	3	0.0153
EEGNetX	3	0.0268
EEGNet	3	0.0518
EEGTCNet	3	0.0175
FBLightConvNet	3	0.0215
ShallowFBCSPNet	3	0.0393
TSception	3	0.0062

the extended Picard algorithm [55], and the components that ICLabel [56] classified as eye, muscle, heart, line noise or channel noise were removed before band-passing to the decoding band. Across the 63 training recordings, a mean of 6.8 of 30 components was removed, most often eye (42 % of removals) and muscle (38 %). Gait-related artefact is difficult to separate from cortical activity during walking [57, 58], and ICLabel is not trained on it specifically, so this control narrows the question and leaves part of it open.

Cleaning changes balanced accuracy from 0.897 to 0.859, a paired difference of -0.038 (higher in 4 participants,

Table 7

Ablation of the reported model on cohort A, three data-split seeds. Rows vary whether the second-order branch is present and where its tangent reference comes from. SD is across seeds; Δ and the test are paired against the stem across participants, with seeds averaged within participant.

Configuration	Acc	SD _{seed}	Δ vs stem	p (higher)
stem alone	0.864	0.032	—	—
stem + branch, running reference	0.909	0.004	+0.045	0.219 (6/7)
stem + branch, reference frozen after fitting	0.825	0.012	−0.039	0.031 (1/7)
stem + branch, no reference	0.843	0.027	−0.021	0.578 (3/7)

Table 8

Artefact control on cohort A, for the reported model. Balanced accuracy of stem + branch on the training task, before and after ICA cleaning, mean over 3 seeds, with the number of muscle components removed per recording. The last row is the mean $\pm$ SD across participants.

Participant	Uncleaned	Cleaned	Change	Muscle removed
sub-01	0.979	0.986	+0.006	2.9
sub-02	0.874	0.880	+0.006	7.1
sub-03	0.991	0.944	−0.047	2.3
sub-04	0.943	0.952	+0.009	1.9
sub-05	0.895	0.929	+0.034	1.3
sub-06	0.774	0.566	−0.207	0.3
sub-07	0.825	0.759	−0.066	2.3
Mean $\pm$ SD	0.897 $\pm$ 0.080	0.859 $\pm$ 0.149	−0.038 $\pm$ 0.082	

lower in 3; Wilcoxon $p = 0.58$; Table 8). Most of the accuracy survives, well above the 0.500 chance level, and the mean is driven by one participant: sub-06 falls by 0.207, and without it the mean change is -0.010 . Four of the seven are unchanged or improve. The change does not follow the amount of muscle activity removed: the participant with the most muscle components removed, 7.1 per recording, changes by $+0.006$, and the Spearman correlation between muscle components removed and accuracy change is $\rho = +0.05$ across the seven participants. Dependence on muscle artefact would produce a negative correlation. Seven participants cannot settle the question, and average re-referencing is part of the cleaning, so the control constrains the artefact account rather than excluding it (Section 8).

Where the reference acts is shown in Fig. 3. The tangent map amplifies the frontocentral midline most (AFz, Fz and F2 at 2.9) and the temporal sites most exposed to jaw and neck muscle least (T7 and T8 at 1.9). The class contrast is separate from that and is a property of the data: absolute 8–30 Hz power falls over left sensorimotor cortex during walking (C3 -3.5 dB, C1 -3.0 , CP3 -2.8), which is the topography and the sign expected of mu and beta desynchronisation, and rises occipitally (O1 and O2 at $+2.1$), which is consistent with optic flow. Gross movement artefact would raise power during walking rather than lower it over sensorimotor cortex, so the sign of the contrast argues against an artefact account without settling it. It is one participant, and a power contrast is not an artefact rejection.

6.3. The running reference and the class-block timescale

The branch maps each window's covariance into the tangent space at a reference $\mathbf{M}$ that is a running mean over the incoming windows. That is what lets the representation follow a session as it changes, and it is also the component that can fail: because $\mathbf{M}$ averages whatever is currently streaming, a protocol that presents one class for longer than the estimator remembers will drive the reference onto that class, and dividing a window of that class by its own covariance removes what separates it from the others.

Condition 1 (Protocol timescale hypothesis). Assume (i) the running estimate is an exponential moving average with nominal memory μ windows; (ii) the within-class covariance is stationary over a block; (iii) the protocol presents classes in contiguous blocks of length τ_{blk} ; and (iv) session drift acts as a congruence transform slow on the scale τ_{drift} . Under (i)–(iv) the estimate both avoids tracking the streaming class and tracks session drift only if

$$\tau_{\text{blk}} < \mu < \tau_{\text{drift}}. \quad (5)$$

Where $\tau_{\text{blk}} \geq \tau_{\text{drift}}$ the band is empty and no setting of μ satisfies both requirements. This is an empirical design hypothesis, not a derived result: we give no proof, its lower side is tested here empirically, and its upper side is not measured.

The lower bound is what this paper measures. It is checked in two ways that do not share a failure mode: across cohorts, by comparing protocols whose block lengths differ by two orders of magnitude (Section 6.5), and within a cohort, by moving the memory directly while everything else is held fixed (Table 12, Fig. 5). The upper bound is not measured here (Section 8). Across a 400-fold range of μ on two cohorts, the sign and the size of the effect track the comparison. On cohort A, where μ exceeds the block length, the branch gains, and driving μ below it reverses the effect to -0.076 . On cohort B, raising μ above the block length takes the cost from 0.175 to 0.046 and then to 0.026, so crossing the boundary bounds the damage without turning it into a gain. The comparison therefore tracks the direction and the severity of the failure; it is not sufficient for a positive effect, and Section 8 says what that leaves open. The row at $N = 8$, $m = 0.2$ separates the two formulations: N has fallen below the block length while μ has not, and the gain survives, so the governing quantity is the memory and not the update size.

6.4. A second-order branch, and where the rate condition sends it

Table 9 reports the branch of Section 4.2 against the stem alone, three seeds everywhere, paired across participants.

On cohort C, which no design decision touched, the branch adds 0.064 and is higher in every one of the 20 participants. Two further readings of that cohort are in Fig. 6. The change exceeds a participant's own standard deviation across the three seeds in 16 of the 20, so for most of them

Table 9

The tangent-space branch against the convolutional stem, three seeds, paired Wilcoxon across participants. Cohort C was held out from architecture and pipeline development. Cohort B is shown at the default adaptation rate, whose 160-window memory is shorter than its 214-window class blocks, and at a memory of 3200 windows, which is far longer than them.

Cohort	Stem	Stem + branch	Δ	Higher	p
A ($n=7$)	0.864	0.909 ± 0.004	+0.045	6/7	0.21875
C ($n=20$), held out	0.779	0.843 ± 0.006	+0.064	20/20	8.8×10^{-5}
B ($n=8$), $m=0.2$	0.744	0.568 ± 0.005	-0.175	0/8	0.00781
B ($n=8$), $m=0.01$	0.744	0.718 ± 0.004	-0.026	3/8	0.3125

the effect is larger than the noise the measurement already carries, and against the eight published decoders on the same cohort the branch leads the strongest of them, EEGNeX, by +0.060, higher in 18 of the 20 ($p = 1.3 \times 10^{-4}$, paired Wilcoxon). It reaches 0.843 against 0.783 for the strongest of the eight published decoders in the same pipeline, and lowers expected calibration error from 0.075 to 0.062. On cohort A it adds 0.045 in 6 of 7 participants. The effect there is not significant at seven participants ($p = 0.22$), and the more robust observation is stability: the seed-to-seed spread of the grand mean falls from 0.032 for the stem to 0.004 with the branch.

Cohort B inverts. At the default rate the branch reaches 0.568 against 0.744, lower in 8 of 8 participants, and sits close to the 0.500 chance line. This is the failure Condition 1 predicts rather than a separate one. That cohort presents Walk in blocks of 214 windows and is 88 % Walk; at $m = 0.2$ the estimator memory is 160 windows, shorter than a block, so the reference converges on the Walk covariance, and whitening a Walk window by the Walk covariance removes the variance that separates the classes. Lengthening the memory to 3200 windows, which the condition prescribes for a 214-window block, returns the cohort to 0.718, indistinguishable from the stem ($p = 0.31$).

Table 10 separates the two roles of the reference. Freezing it at the fitting-split mean removes the ability to track the streaming class but not the class imbalance in the fitting data itself, and recovers cohort B only partly. Removing it altogether repairs cohort B and is null on cohort C and negative on cohort A, so the adapting reference is where the gain comes from and the no-reference variant is a cohort B patch with no support elsewhere. We therefore retain the adapting reference and let Condition 1, read from the protocol, decide where the branch is admissible. Nothing in that choice is fitted on a confirmation cohort.

Inside one trained model, the fused representation separates the classes more than either path does. Training one cohort A participant with the reported settings and recording activations with forward hooks on held-out windows (Fig. 2, held-out balanced accuracy 0.967), class separation measured as the mean standardised difference per unit is 0.74 at the stem and 1.22 at the branch, and 2.41 after the two are fused. This is one participant and illustrates the fusion; it is not a population result. A decoder built on the branch alone was not trained, so the paper makes no claim about

Table 10

Where the tangent reference comes from, three seeds, Δ against the stem alone on the same cohort. The adapting reference is the only setting that helps on the two cohorts whose class blocks are short.

Reference	Cohort A	Cohort B	Cohort C
Running (adapting)	+0.045	-0.175	+0.064
Frozen at fitting mean	-0.039	-0.026	-
None (no whitening)	-0.021	+0.011	+0.003

the branch alone against the stem; the population result is that the fusion exceeds the stem (Table 11).

6.5. Five cohorts, and one boundary that separates them

Table 11 collects the branch against the convolutional stem on every cohort, three seeds throughout, paired across participants. Cohorts D and E were added after the method was fixed, and nothing was tuned on either. Both axes of Fig. 7(b) are fixed before training: the block length is read from the labels and the memory is fixed by the optimiser settings, so the separation uses no accuracy. The regime column of Table 11 follows from that comparison alone, for the same reason.

The branch helps on four cohorts and fails on one. The cohort it fails on is the only one whose class blocks outlast the estimator memory, and that comparison is available before any model is trained: τ_{blk} comes from the label sequence and $\mu = N/m$ from the batch size and momentum. On cohorts D, C, E and A a block is 1, 5, 13 and 18 windows against a memory of 160, so the reference averages over many blocks. The quantity is the memory and not the update size alone: on cohort A an update of 8 windows at $m = 0.2$ keeps the gain, and the same update at $m = 1.0$, a memory of 8, removes it (Table 12). On cohort B a block is 214 windows, longer than the 160-window memory, and the reference converges on the Walk covariance; whitening a Walk window by the Walk covariance removes the variance separating the classes, and accuracy falls to 0.568 against a 0.500 chance line.

Lengthening the estimator memory on cohort B moves it back monotonically: 0.568 at a memory of 160 windows, 0.650 at 428 and 0.718 at 3200, against 0.744 for the stem alone. The direction the comparison predicts is visible at every point measured.

Two limits belong with this claim. The boundary was identified after cohorts A, B and C were run, so only its applications to cohorts D and E were prospective. And the boundary itself is untested: every cohort where the branch helps has τ_{blk} between 1 and 18, the one where it fails has 214, and no cohort has a block between 32 and 160 windows, which is why the block length appeared for a time to compete with N rather than with μ . Table 12 settles it by the experiment that would test it directly.

Table 11

The tangent-space branch against the stem alone, three seeds, paired Wilcoxon across participants with seeds averaged. Each cohort is a separate test of the same hypothesis, so p is per cohort and uncorrected (Section 5.4); across the four cohorts in the safe regime, 40 of the 43 participants improve. τ_{blk} is the median single-class block in windows, read from the fitting labels before training, and the estimator memory is $\mu = N/m = 160$ windows at the settings used throughout.

Cohort	n	τ_{blk}	Regime	Stem	Stem + branch	Δ	Higher	p
C, motor execution	20	5	safe	0.779	0.843	+0.064	20/20	8.8×10^{-5}
D, BCI IV-2a	9	1	safe	0.701	0.741	+0.040	8/9	0.020
E, exoskeleton (DECODED)	7	13	safe	0.928	0.949	+0.021	6/7	0.047
A, exoskeleton	7	18	safe	0.864	0.909	+0.045	6/7	0.219
B, treadmill	8	214	at risk	0.744	0.568	-0.175	0/8	0.008

Table 12

The estimator memory against the class-block length, varied directly. $\mu = N/m$ is the memory in windows, set by the number N averaged into one covariance update and the momentum m . Controls are matched on N , which changes training, while m is inert for a stem that maintains no running statistic.

Cohort	τ_{blk}	N	m	μ	$\mu > \tau_{\text{blk}}$	Δ	Higher	p
A	18	8	1	8	no	-0.076	0/7	0.016
A	18	8	0.2	40	yes	+0.048	6/7	0.156
A	18	32	0.2	160	yes	+0.045	6/7	0.219
B	214	32	0.2	160	no	-0.175	0/8	0.008
B	214	256	0.2	1280	yes	-0.046	3/8	0.109
B	214	32	0.01	3200	yes	-0.026	3/8	0.312

Table 13

The quantities a controller would act on, all five cohorts, three seeds, mean over participants. ECE is expected calibration error. FA/min counts spurious transitions into the positive class per minute of the negative class, and Missed is the fraction of true onsets never detected, both defined in Section 5.3. On cohorts A, B and E the positive class is walking and these read as spurious Walk commands per minute of standing; cohorts C and D are rest against movement and left hand against right hand, where the same arithmetic applies but no walk command exists, so the generic reading is the correct one there. Both are computed on the raw decision sequence, with no smoothing, hysteresis or safety logic, so they are proxies for device behaviour rather than measurements of it.

Cohort	Model	Acc	ECE	FA/min	Missed
A, exoskeleton	stem	0.864	0.063	0.70	0.085
	+ branch	0.909	0.049	2.09	0.028
B, treadmill	stem	0.744	0.066	7.56	0.000
	+ branch	0.568	0.051	6.26	0.000
C, motor execution	stem	0.779	0.075	6.59	0.072
	+ branch	0.843	0.062	5.25	0.026
D, BCI IV-2a	stem	0.701	0.056	12.10	0.294
	+ branch	0.741	0.068	13.40	0.199
E, exoskeleton	stem	0.928	0.043	1.80	0.000
	+ branch	0.949	0.037	1.46	0.000

6.6. What the branch costs a controller

Table 13 reports the quantities a controller would act on. Calibration improves with the branch on 4 of the five cohorts and the missed-onset rate falls on 3, so the accuracy gain is not bought by a worse-behaved posterior. The exception is the spurious-activation rate on cohort A, which rises from

Table 14

Inference cost at cohort A's input shape. The benchmark passes eight windows per forward and returns one decision, more than the reported model needs, since it classifies each window on its own, so the figures are an upper bound on per-window cost. Latency is the median over 200 forwards after warm-up, batch 1, with CUDA synchronised; the 95th percentile is given for the online case. No training.

	Stem	Stem + branch	Ratio
Parameters	20,019	290,943	14.5×
GFLOPs per forward	0.32	0.36	1.13×
CPU latency (ms)	5.8	10.0	1.74×
CPU 95th percentile (ms)	7.2	12.0	
GPU latency (ms)	0.64	4.31	6.7×
GPU peak memory (MB)	60	61	

0.70 to 2.09 per minute of standing. That rate is the quantity a per-window error rate hides, since one long false run and many scattered false windows score identically. A decoder that is more often right per window can still command more false starts, because the two are different functions of the same posterior, and on a lower-limb exoskeleton the second is the one that throws a wearer off balance. We report it rather than the accuracy alone.

6.7. What the branch costs at inference

Parameters are the wrong measure of what the branch costs to run. Its expense is an eigendecomposition of a 60×60 matrix, which no parameter count reflects and which a floating-point count misses as well. Table 14 measures the model instead, at cohort A's input shape, with no training.

The decision rate sets the budget. Cohort A advances one window every 0.5 s, and one forward covering eight windows takes 10.0 ms on a CPU, with a 95th percentile of 12.0 ms. That is around 2 % of the interval, so the reported model runs in the loop on a CPU and needs no accelerator. The label-free reference update, the part with no analogue in the stem, is 0.98 ms of that.

Two things in the table are worth stating because they cut in opposite directions. The branch multiplies the parameter count by 14.5 but the CPU latency by only 1.74, so parameters overstate what it costs to run. Floating-point operations understate it, rising by 1.13×, because the counter does not see the eigendecomposition. Measured latency is therefore

Table 15

Published decoders on the two cohorts added last, mean and standard deviation over three seeds, same pipeline as Table 3. † marks a decoder left near chance for at least five of the nine cohort D participants on at least one seed.

Decoder	Cohort D	Cohort E
ShallowFBCSPNet	0.682 $\pm$ 0.013	0.886 $\pm$ 0.008
FBLightConvNet	0.674 $\pm$ 0.004	0.891 $\pm$ 0.022
EEGNetX	0.656 $\pm$ 0.015	0.887 $\pm$ 0.018
EEGConformer†	0.599 $\pm$ 0.077	0.896 $\pm$ 0.010
TSception†	0.579 $\pm$ 0.015	0.833 $\pm$ 0.011
EEGNet†	0.526 $\pm$ 0.017	0.700 $\pm$ 0.054
Deep4Net†	0.525 $\pm$ 0.007	0.722 $\pm$ 0.035
EEGTCNet†	0.505 $\pm$ 0.015	0.695 $\pm$ 0.016
Stem alone	0.701	0.928
Stem + branch	0.741	0.949

the more informative estimate of computational cost. On the GPU the overhead is proportionally larger, 6.7 $\times$, because an eigendecomposition of a small matrix is latency-bound and the device cannot hide it behind throughput.

6.8. Published decoders on the two new cohorts

Table 15 places the branch against the eight published decoders on the two cohorts added last. All are trained in the same pipeline with the same splits, seeds and metric code as on cohorts A, B and C, so the comparison is within-harness. The branch leads every published decoder on both: 0.741 against 0.682 for the strongest on cohort D, and 0.949 against 0.896 on cohort E. The stem alone also clears them, so the ordering does not depend on the branch; what the branch adds is the margin. Cohort D needs a caveat the margin itself hides. Five of its eight decoders are left near chance for most participants on at least one seed, which the roughly 100 fitting windows per participant explain, so the wide gaps there measure the fragility of those decoders under shared settings rather than anything about the branch. Paired over the nine participants with Holm correction across the eight comparisons, the three that trained give +0.059 over ShallowFBCSPNet (higher in 7 of 9, $p = 0.039$), +0.085 over EEGNetX (8 of 9, $p = 0.039$) and +0.067 over FBLightConvNet (9 of 9, $p = 0.031$). Those are the comparisons cohort D actually supports, and all three hold after correction. The dagger marks a fragile regime rather than a fixed property of a decoder: running the identical criterion on an independent repeat of these cells on a second machine moves EEGNetX across the threshold, at four near-chance participants here and five there.

6.9. The reported model against the field

Table 3 is the comparison the rest of the results build toward: the reported model, the stem it is built on, and eight published decoders, on every cohort, three seeds each. Every cell was trained in the same pipeline with the same splits, seeds, preprocessing and metric code, so the columns are internally comparable. They are not comparable with numbers from these decoders' original papers, which used

different splits and different preprocessing, and we make no claim about those.

The reported model is highest on 4 of the five cohorts. On the fifth, cohort B, it is the worst entry in the column, and that is exactly the cohort where a class block outlasts the estimator memory. What makes this a usable result rather than an unreliable one is that the stem alone leads that column: 0.744 against 0.734 for the strongest published decoder there. One of the two configurations is therefore the best entry in every column of the table, and the block-length condition selects between them from the recording protocol, before any data is collected and without consulting accuracy.

Parameter counts belong beside this. The reported model is 290,943 parameters against 20,019 for the stem as a runnable model, so the gains in this table cost roughly fifteen times the parameters. Of the published decoders, ShallowFBCSPNet and EEGNet are an order of magnitude smaller again, and EEG Conformer is larger than ours. Nothing here is a claim about efficiency.

One comparison in this table is with neural decoders only, and the branch reads a Riemannian representation, so the obvious question is whether the classical pipeline would do as well without a network. On cohort A, in this harness and on these splits, it does not. Tangent space at the Riemannian mean of the fitting covariances followed by logistic regression reaches 0.793 over three seeds, and minimum distance to the Riemannian mean reaches 0.706, against 0.909 for the reported model: a paired difference of +0.116, higher in 7 of the 7 participants ($p = 0.016$). Expected calibration error is also worse, 0.078 against 0.049. The classical pipeline sits below the convolutional stem alone as well, at 0.864, though with seven participants that gap is not significant. Its reference is the Riemannian mean of the fitting split, fitted once and then held fixed, which is the static counterpart of the running estimate Section 6.3 studies.

On cohort C the outcome is different. Tangent space with logistic regression reaches 0.840 $\pm$ 0.005 over three seeds against 0.843 for the reported model, a paired difference of +0.003, higher in 11 of the 20 participants ($p = 0.52$), with calibration error 0.067 against 0.062; minimum distance to the Riemannian mean reaches 0.701. The classical pipeline also exceeds the convolutional stem alone on this cohort, by 0.061 (higher in 18 of 20, $p = 2.2 \times 10^{-4}$). On the held-out cohort, then, the branch brings the network level with a classical tangent-space classifier and no further. The two cohorts differ in what a fixed reference has to bridge: cohort C is evaluated on later runs of the same recording protocol as its fitting runs, with no interval between them reported, while cohort A is evaluated on sessions recorded weeks after its fitting sessions, and a reference fitted once and then held fixed is the component expected to degrade across such an interval. That reading is consistent with both results; it is not tested separately here.

7. Discussion

7.1. A distribution-level improvement need not be a task-level one

Domain adaptation and invariance methods are routinely justified by a distribution-level quantity, a divergence, an alignment distance or a nuisance-decodability score, with task performance assumed to follow [59–62]. The reference ablation of Section 6.5 is a counterexample within this system. Removing the reference entirely makes the representation strictly less adapted to the incoming session, and on the cohort whose protocol violates the rate condition it is the variant that works; on the two cohorts that satisfy it, the same change is null or negative. Whether an adaptive statistic helps is therefore a property of the protocol it adapts on, not of how much adaptation it performs.

7.2. Relation to temporally correlated test-time adaptation

The vision literature has identified that test-time adaptation degrades when the test stream is temporally correlated [21–23], and proposes reservoirs and stabilised objectives as remedies. Our result is compatible with that line and adds two things specific to the blocked-protocol case. First, the relevant quantity is a ratio of timescales, measured on the stream, and which side of the threshold a protocol falls on can be predicted in advance, as the block-length comparison shows (Section 6.5). Second, slowing adaptation is not free. On both cohorts with short blocks it costs accuracy, so remedies that slow adaptation can exchange class tracking for a different loss, which on these cohorts could not be attributed to drift tracking directly.

8. Limitations

Several limitations constrain the interpretation of these results.

Seeds and effect size. Every cohort in Table 3 is three data-split seeds. The effect of the branch is nevertheless modest in absolute terms, +0.021 to +0.064 where it helps, and on cohort A the paired bootstrap interval spans zero ($[-0.020, +0.109]$, $p = 0.22$ at seven participants). The cohort that carries the claim most strongly is C, with twenty participants, all twenty improving ($p = 8.8 \times 10^{-5}$). Cohorts D and E are smaller replications, each significant on its own test ($p = 0.020$ and 0.047) with bootstrap intervals that exclude zero, and across the four cohorts in the safe regime 40 of the 43 participants improve. A reader who wants a single number should take cohort C's +0.064 and treat cohort A as consistent with it rather than as independent support.

No paired significance tests on accuracy. Per-participant accuracies were not retained for the single-seed component ablation and the single-seed rate tables, so those comparisons carry dispersion without paired tests. Paired tests are reported wherever per-participant rows exist: the temporal and normalisation controls, the artefact control (Section 6.2), the contrasts against the stem on cohort A

(Section 6.1), and the comparison with published decoders on cohort C (Section 6.9).

Cohort size. Seven, eight, twenty, nine and seven participants, 51 in total. This is typical for exoskeleton BCI studies [1] but small, and the between-subject SD of 0.06 to 0.13 is large relative to the differences between arms. **A classical classifier matches the reported model on the held-out cohort.** Tangent space with logistic regression and MDM were run on cohorts A and C (Section 6.9). The reported model exceeds the tangent-space classifier by 0.116 on cohort A and is level with it on cohort C (+0.003, $p = 0.52$). Its accuracy advantage over the classical pipeline is therefore established only on the longitudinal cohort, and on the held-out cohort the network's gain over its own stem is matched by a model with no network.

Best-epoch evaluation. Model selection uses the best epoch by held-out balanced accuracy, which is conventional but upward-biased on small folds. The bias applies equally to all arms and baselines within a pipeline, so comparisons within a pipeline are unaffected, but absolute values should be read as optimistic.

Baseline training. The published decoders were trained with the proposed decoder's settings and no per-model tuning. EEGNet, Deep4Net and EEG-TCNet stayed near chance for several cohort C participants under those settings, so their cohort C results record a training failure and are not evidence about those architectures.

The upper bound is not measured. Two label-free estimates of τ_{drift} were attempted. Block-wise covariance distances within a session are dominated by class alternation where blocks are short, and on cohort A they return about 18 windows, the class-block length. Distances between session mean covariances are already at their plateau at a gap of one session for every cohort A participant, and fall with the gap for two, so a fitted timescale lies below the sampling interval and is not a measurement; cohort B's three trials give two gaps, too few for a fit. The data therefore bound the drift timescale only from above, at about one session, which does not constrain the adaptation rate. A protocol with more closely spaced recordings is needed to test the upper bound.

Condition 1 is hypothesis-generating. It was formulated after observing cohort B fail. One prediction from it has since been tested out-of-sample, and that test falsified the one-sided form, which is how the two-sided form arose. A stronger prescription, setting the adaptation rate from the block length, was also falsified (Section 6.5). The two-sided form has been applied prospectively only to cohorts D and E, and its upper bound has not been tested in advance.

Artefact. Retraining the reported model after ICA cleaning changes balanced accuracy by -0.038 (Wilcoxon $p = 0.58$; four of seven participants unchanged or better), and the mean is carried by sub-06 alone, which falls by 0.207 (Section 6.2). The change does not follow the amount of muscle activity removed ($\rho = +0.05$). The band-power contrast points the same way: power falls over sensorimotor cortex during walking (Fig. 3c), whereas gross movement artefact would raise it. Neither result establishes that the

accuracy is neural. Seven participants are too few, average re-referencing is part of the cleaning, ICLabel is not trained on gait-related artefact specifically, and a power contrast in one participant is not an artefact rejection.

Block length depends on windowing. τ_{blk} and the memory are both counted in windows, so for a given protocol they depend on the window stride as well: the same 10 s block is 20 windows at a 0.5 s stride and 10 at 1 s. Window length and stride differ between cohorts (Table 2), so each comparison is made within one cohort's own windowing, and a statement in seconds would need both quantities converted together.

Parameter cost. The branch is 238,028 parameters and the fusion it requires a further 32,896, against 19,505 in the stem path and 514 in the shared classifier. The reported model is therefore about fifteen times the size of the stem it improves on, for 0.02 to 0.06 of balanced accuracy. Two of the published decoders it beats are an order of magnitude smaller again. Whether the tangent vector can be projected more cheaply, for instance by reducing its dimension before the linear layer, is not tested here and is the obvious first place to look.

The block-length boundary was found after the fact. The comparison between τ_{blk} and the estimator memory separates all five cohorts with nothing fitted, but it was identified after cohorts A, B and C had been run. Only its applications to cohorts D and E were prospective. The boundary is supported, not established, and the test that would establish it is a cohort chosen in advance because its protocol sits near the boundary.

The boundary is untested at the boundary. Across cohorts the boundary is straddled but not located: every cohort where the branch helps has a block of 1 to 18 windows and the one where it fails has 214, against a nominal memory of 160. Nothing is measured between 18 and 214, so across cohorts the location of the change is inferred from five points that straddle it. Within cohorts it is measured directly: varying μ at fixed block length crosses the boundary in both directions (Table 12).

No calibrated threshold. Two manipulations move the effect in the predicted direction and neither restores parity. Setting the adaptation rate from the block length with a fixed margin of two was pre-specified in version control and falsified. Enlarging the update to 256 windows against a 214-window block recovers 74 % of the loss and still costs 0.046. The condition therefore gives an ordering and a direction, not a rate a practitioner could set, and the upper side of the two-sided condition remains unmeasured on both cohorts where it was attempted.

Cohort E tests task, not drift. Only one participant recorded on two dates, so its split is temporal across runs within a session. It is the closest match in the paper to the target application and the weakest test of longitudinal shift, and those two facts are in tension.

Reproducibility across machines. The cohort D baseline sweep was run independently on two machines. Sixteen of 24 cells differ, by up to 0.022 on a single cell, while the

three-seed arm means agree to 0.003 and the strongest published decoder is identical on both. Single-cell numbers anywhere in this paper therefore carry cross-machine variation wider than several of the gaps between neighbouring rows, and only seed-averaged quantities should be compared.

9. Conclusion

A log-power stem sees the covariance between channels only through its learned spatial filters. Supplying that structure explicitly through a tangent-space branch raises balanced accuracy on four of five cohorts and 51 participants, with the highest mean accuracy among eight published decoders trained untuned in the same pipeline. It fails on the fifth, and a quantity read from the protocol separates the two cases: the branch's reference is an average over whatever is currently streaming, so it survives only while its memory outlasts a class block.

That condition is the part worth building on, and it is also the part least finished. It separates five cohorts with nothing fitted, it predicted the sign on two cohorts prospectively, and enlarging the update recovers most of the loss where it is violated. It still yields no rate a practitioner could set, its upper side has never been measured, and no cohort has been run near the boundary itself. The experiment that would settle it is small and specific: choose a protocol whose class blocks fall between 18 and 214 windows, register the prediction, and run it once.

The wider point is that a label-free adaptive statistic inherits the structure of the protocol it adapts on. Test-time adaptation methods are usually evaluated on streams that are shuffled or assumed i.i.d.; EEG protocols are blocked by design, and the block length is known before a single recording is made. Reporting it alongside accuracy would let a reader see, without running anything, whether an adaptive component is operating inside its domain of validity.

CRedit authorship contribution statement

Esm E Moula Chowdhury Abha: Software, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Md Wahiduzzaman Suva:** Software, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Muhammad Hasibur Rashid Chayon:** Supervision, Project administration, Writing – review and editing.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All five datasets are publicly available. All code, configurations and result files, including those of

experiments that failed, are in the accompanying repository at <https://github.com/wshuv-o/calmmnet>, on branch representation-beats-architecture at tag manuscript-v1, which is the state from which every number in this paper was produced.

References

- [1] Y. He, D. Eguren, J. M. Azorín, R. G. Grossman, T. P. Luu, J. L. Contreras-Vidal, Brain-machine interfaces for controlling lower-limb powered robotic systems, *Journal of Neural Engineering* 15 (2018) 021004. URL: <https://doi.org/10.1088/1741-2552/aaa8c0>. doi:10.1088/1741-2552/aaa8c0.
- [2] E. López-Larraz, F. Trincado-Alonso, V. Rajasekaran, S. Peres-Nombela, A. J. del Ama, J. Aranda, J. Minguez, A. Gil-Agudo, L. Montesano, Control of an ambulatory exoskeleton with a brain-machine interface for spinal cord injury gait rehabilitation, *Frontiers in Neuroscience* 10 (2016) 359. URL: <https://doi.org/10.3389/fnins.2016.00359>. doi:10.3389/fnins.2016.00359.
- [3] A. H. Do, P. T. Wang, C. E. King, S. N. Chun, Z. Nenadic, Brain-computer interface controlled robotic gait orthosis, *Journal of NeuroEngineering and Rehabilitation* 10 (2013) 111. URL: <https://doi.org/10.1186/1743-0003-10-111>. doi:10.1186/1743-0003-10-111.
- [4] J. L. Contreras-Vidal, R. G. Grossman, NeuroRex: a clinical neural interface roadmap for EEG-based brain machine interfaces to a lower body robotic exoskeleton, *Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (2013) 1579–1582. URL: <https://doi.org/10.1109/embc.2013.6609816>. doi:10.1109/embc.2013.6609816.
- [5] A. Presacco, R. Goodman, L. Forrester, J. L. Contreras-Vidal, Neural decoding of treadmill walking from noninvasive electroencephalographic signals, *Journal of Neurophysiology* 106 (2011) 1875–1887. URL: <https://doi.org/10.1152/jn.00104.2011>. doi:10.1152/jn.00104.2011.
- [6] P. Shenoy, M. Krauledat, B. Blankertz, R. P. N. Rao, K.-R. Müller, Towards adaptive classification for BCI, *Journal of Neural Engineering* 3 (2006) R13–R23. URL: <https://doi.org/10.1088/1741-2560/3/1/r02>. doi:10.1088/1741-2560/3/1/r02.
- [7] M. Krauledat, M. Tangermann, B. Blankertz, K.-R. Müller, Towards zero training for brain-computer interfacing, *PLoS ONE* 3 (2008) e2967. URL: <https://doi.org/10.1371/journal.pone.0002967>. doi:10.1371/journal.pone.0002967.
- [8] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, K.-R. Müller, Optimizing spatial filters for robust EEG single-trial analysis, *IEEE Signal Processing Magazine* 25 (2008) 41–56. URL: <https://doi.org/10.1109/msp.2008.4408441>. doi:10.1109/msp.2008.4408441.
- [9] C. Vidaurre, C. Sannelli, K.-R. Müller, B. Blankertz, Machine-learning-based coadaptive calibration for brain-computer interfaces, *Neural Computation* 23 (2011) 791–816. URL: https://doi.org/10.1162/neco_a_00089. doi:10.1162/neco_a_00089.
- [10] R. T. Schirrmester, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggenberger, M. Tangermann, F. Hutter, W. Burgard, T. Ball, Deep learning with convolutional neural networks for EEG decoding and visualization, *Human Brain Mapping* 38 (2017) 5391–5420. URL: <https://doi.org/10.1002/hbm.23730>. doi:10.1002/hbm.23730.
- [11] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, B. J. Lance, EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces, *Journal of Neural Engineering* 15 (2018) 056013. URL: <https://doi.org/10.1088/1741-2552/aace8c>. doi:10.1088/1741-2552/aace8c.
- [12] A. Barachant, S. Bonnet, M. Congedo, C. Jutten, Multiclass brain-computer interface classification by Riemannian geometry, *IEEE Transactions on Biomedical Engineering* 59 (2012) 920–928. URL: <https://doi.org/10.1109/tbme.2011.2172210>. doi:10.1109/tbme.2011.2172210.
- [13] M. Congedo, A. Barachant, R. Bhatia, Riemannian geometry for EEG-based brain-computer interfaces: a primer and a review, *Brain-Computer Interfaces* 4 (2017) 155–174. URL: <https://doi.org/10.1080/2326263x.2017.1297192>. doi:10.1080/2326263x.2017.1297192.
- [14] H. He, D. Wu, Transfer learning for brain-computer interfaces: a Euclidean space data alignment approach, *IEEE Transactions on Biomedical Engineering* 67 (2020) 399–410. URL: <https://doi.org/10.1109/tbme.2019.2913914>. doi:10.1109/tbme.2019.2913914.
- [15] P. L. C. Rodrigues, C. Jutten, M. Congedo, Riemannian Procrustes analysis: transfer learning for brain-computer interfaces, *IEEE Transactions on Biomedical Engineering* 66 (2019) 2390–2401. URL: <https://doi.org/10.1109/tbme.2018.2889705>. doi:10.1109/tbme.2018.2889705.
- [16] O. Yair, M. Ben-Chen, R. Talmon, Parallel transport on the cone manifold of SPD matrices for domain adaptation, *IEEE Transactions on Signal Processing* 67 (2019) 1797–1811. URL: <https://doi.org/10.1109/tsp.2019.2894801>. doi:10.1109/tsp.2019.2894801.
- [17] P. Zanini, M. Congedo, C. Jutten, S. Said, Y. Berthoumieu, Transfer learning: a Riemannian geometry framework with applications to brain-computer interfaces, *IEEE Transactions on Biomedical Engineering* 65 (2018) 1107–1116. URL: <https://doi.org/10.1109/tbme.2017.2742541>. doi:10.1109/tbme.2017.2742541.
- [18] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, T. Darrell, Tent: fully test-time adaptation by entropy minimization, in: *International Conference on Learning Representations (ICLR)*, 2021. URL: <https://arxiv.org/abs/2006.10726>.
- [19] S. Schneider, E. Rusak, L. Eck, O. Bringmann, W. Brendel, M. Bethge, Improving robustness against common corruptions by covariate shift adaptation, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL: <https://arxiv.org/abs/2006.16971>.
- [20] Z. Nado, S. Padhy, D. Sculley, A. D'Amour, B. Lakshminarayanan, J. Snoek, Evaluating prediction-time batch normalization for robustness under covariate shift, *arXiv preprint arXiv:2006.10963* (2020). URL: <https://arxiv.org/abs/2006.10963>.
- [21] T. Gong, J. Jeong, T. Kim, Y. Kim, J. Shin, S.-J. Lee, NOTE: robust continual test-time adaptation against temporal correlation, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL: <https://doi.org/10.52202/068431-1976>. doi:10.52202/068431-1976.
- [22] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, M. Tan, Towards stable test-time adaptation in dynamic wild world, in: *International Conference on Learning Representations (ICLR)*, 2023. URL: <https://arxiv.org/abs/2302.12400>.
- [23] Q. Wang, O. Fink, L. Van Gool, D. Dai, Continual test-time domain adaptation, in: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. URL: <https://doi.org/10.1109/cvpr52688.2022.00706>. doi:10.1109/cvpr52688.2022.00706.
- [24] X. Chen, X. Teng, H. Chen, Y. Pan, P. Geyer, Toward reliable signals decoding for electroencephalogram: a benchmark study to EEGNeX, *Biomedical Signal Processing and Control* 87 (2024) 105475. URL: <https://doi.org/10.1016/j.bspc.2023.105475>. doi:10.1016/j.bspc.2023.105475.
- [25] Y. Roy, H. Banville, I. Albuquerque, A. Gramfort, T. H. Falk, J. Faubert, Deep learning-based electroencephalography analysis: a systematic review, *Journal of Neural Engineering* 16 (2019) 051001. URL: <https://doi.org/10.1088/1741-2552/ab260c>. doi:10.1088/1741-2552/ab260c.
- [26] A. Craik, Y. He, J. L. Contreras-Vidal, Deep learning for electroencephalogram (EEG) classification tasks: a review, *Journal of Neural Engineering* 16 (2019) 031001. URL: <https://doi.org/10.1088/1741-2552/ab0ab5>. doi:10.1088/1741-2552/ab0ab5.
- [27] Y. Song, Q. Zheng, B. Liu, X. Gao, EEG Conformer: convolutional transformer for EEG decoding and visualization, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 31 (2023) 710–719. URL: <https://doi.org/10.1109/tnsre.2022.3230250>. doi:10.1109/tnsre.2022.3230250.

- [28] T. M. Ingolfsson, M. Hersche, X. Wang, N. Kobayashi, L. Cavigelli, L. Benini, EEG-TCNet: an accurate temporal convolutional network for embedded motor-imagery brain-machine interfaces, in: IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2020. URL: <https://doi.org/10.1109/smc42975.2020.9283028>. doi:10.1109/smc42975.2020.9283028.
- [29] A. Barachant, S. Bonnet, M. Congedo, C. Jutten, Classification of covariance matrices using a Riemannian-based kernel for BCI applications, *Neurocomputing* 112 (2013) 172–178. URL: <https://doi.org/10.1016/j.neucom.2012.12.039>. doi:10.1016/j.neucom.2012.12.039.
- [30] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, F. Yger, A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update, *Journal of Neural Engineering* 15 (2018) 031005. URL: <https://doi.org/10.1088/1741-2552/aab2f2>. doi:10.1088/1741-2552/aab2f2.
- [31] D. Wu, Y. Xu, B.-L. Lu, Transfer learning for EEG-based brain-computer interfaces: a review of progress made since 2016, *IEEE Transactions on Cognitive and Developmental Systems* 14 (2022) 4–19. URL: <https://doi.org/10.1109/tcds.2020.3007453>. doi:10.1109/tcds.2020.3007453.
- [32] L. Haxel, J. Kapoor, U. Ziemann, J. H. Macke, EDAPT: towards calibration-free BCIs with continual online adaptation, *Journal of Neural Engineering* 23 (2026). URL: <https://doi.org/10.1088/1741-2552/ae5689>. doi:10.1088/1741-2552/ae5689.
- [33] Z. Huang, L. Van Gool, A Riemannian network for SPD matrix learning, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 31, 2017.
- [34] Y. Sun, X. Wang, Z. Liu, J. Miller, A. A. Efros, M. Hardt, Test-time training with self-supervision for generalization under distribution shifts, in: Proceedings of the 37th International Conference on Machine Learning (ICML), 2020. URL: <https://www.semanticscholar.org/paper/a436e41141a9d9c5dfc44d16fade67b9a3b76778>.
- [35] M. Boudiaf, R. Mueller, I. Ben Ayed, L. Bertinetto, Parameter-free on-line test-time adaptation, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022. URL: <https://doi.org/10.1109/cvpr52688.2022.00816>. doi:10.1109/cvpr52688.2022.00816.
- [36] J. T. Gwin, K. Gramann, S. Makeig, D. P. Ferris, Electrocortical activity is coupled to gait cycle phase during treadmill walking, *NeuroImage* 54 (2011) 1289–1296. URL: <https://doi.org/10.1016/j.neuroimage.2010.08.066>. doi:10.1016/j.neuroimage.2010.08.066.
- [37] M. Seeber, R. Scherer, J. Wagner, T. Solis-Escalante, G. R. Müller-Putz, EEG beta suppression and low gamma modulation are different elements of human upright walking, *Frontiers in Human Neuroscience* 8 (2014) 485. URL: <https://doi.org/10.3389/fnhum.2014.00485>. doi:10.3389/fnhum.2014.00485.
- [38] T. P. Luu, S. Nakagome, Y. He, J. L. Contreras-Vidal, Real-time EEG-based brain-computer interface to a virtual avatar enhances cortical involvement in human treadmill walking, *Scientific Reports* 7 (2017) 8895. URL: <https://doi.org/10.1038/s41598-017-09187-0>. doi:10.1038/s41598-017-09187-0.
- [39] J. Wagner, T. Solis-Escalante, P. Grieshofer, C. Neuper, G. Müller-Putz, R. Scherer, Level of participation in robotic-assisted treadmill walking modulates midline sensorimotor EEG rhythms in able-bodied subjects, *NeuroImage* 63 (2012) 1203–1211. URL: <https://doi.org/10.1016/j.neuroimage.2012.08.019>. doi:10.1016/j.neuroimage.2012.08.019.
- [40] G. Pfurtscheller, F. H. Lopes da Silva, Event-related EEG/MEG synchronization and desynchronization: basic principles, *Clinical Neurophysiology* 110 (1999) 1842–1857. URL: [https://doi.org/10.1016/s1388-2457\(99\)00141-8](https://doi.org/10.1016/s1388-2457(99)00141-8). doi:10.1016/s1388-2457(99)00141-8.
- [41] G. Pfurtscheller, C. Brunner, A. Schlögl, F. H. Lopes da Silva, Mu rhythm (de)synchronization and EEG single-trial classification of different motor imagery tasks, *NeuroImage* 31 (2006) 153–159. URL: <https://doi.org/10.1016/j.neuroimage.2005.12.003>. doi:10.1016/j.neuroimage.2005.12.003.
- [42] M. Sugiyama, M. Krauledat, K.-R. Müller, Covariate shift adaptation by importance weighted cross validation, *Journal of Machine Learning Research* 8 (2007) 985–1005. URL: <https://doi.org/10.5555/1314498.1390324>. doi:10.5555/1314498.1390324.
- [43] G. Schalk, D. McFarland, T. Hinterberger, N. Birbaumer, J. Wolpaw, BCI2000: A general-purpose brain-computer interface (BCI) system, *IEEE Transactions on Biomedical Engineering* 51 (2004) 1034–1043. URL: <https://doi.org/10.1109/tbme.2004.827072>. doi:10.1109/tbme.2004.827072.
- [44] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, H. E. Stanley, Physiobank, PhysioToolkit, and physionet, *Circulation* 101 (2000). URL: <https://doi.org/10.1161/01.cir.101.23.e215>. doi:10.1161/01.cir.101.23.e215.
- [45] M. Tangermann, K.-R. Müller, A. Aertsen, N. Birbaumer, C. Braun, C. Brunner, R. Leeb, C. Mehring, K. J. Miller, G. R. Müller-Putz, G. Nolte, G. Pfurtscheller, H. Preissl, G. Schalk, A. Schlögl, C. Vidaurre, S. Waldert, B. Blankertz, Review of the BCI competition IV, *Frontiers in Neuroscience* 6 (2012) 55. URL: <https://doi.org/10.3389/fnins.2012.00055>. doi:10.3389/fnins.2012.00055.
- [46] M. Ortiz, L. de la Ossa, J. Juan, E. Iáñez, D. Torricelli, J. Tornero, J. M. Azorín, An EEG database for the cognitive assessment of motor imagery during walking with a lower-limb exoskeleton, *Scientific Data* 10 (2023). URL: <https://doi.org/10.1038/s41597-023-02243-7>. doi:10.1038/s41597-023-02243-7.
- [47] A. Gramfort, M. Luessi, E. Larson, D. A. Engemann, D. Strohmeier, C. Brodbeck, R. Goj, M. Jas, T. Brooks, L. Parkkonen, M. Hämäläinen, MEG and EEG data analysis with MNE-Python, *Frontiers in Neuroscience* 7 (2013) 267. URL: <https://doi.org/10.3389/fnins.2013.00267>. doi:10.3389/fnins.2013.00267.
- [48] I. Loshchilov, F. Hutter, Decoupled weight decay regularization, in: International Conference on Learning Representations (ICLR), 2019. URL: <https://openreview.net/forum?id=Bkg6RiCqY7>.
- [49] R. T. Schirrmester, A. Gramfort, S. Chevallier, et al., Braindecode: a deep learning toolbox to decode raw time-domain EEG, 2024. URL: <https://braindecode.org>, software. Model implementations used as published; the method reference for the convolutional decoders is Schirrmester et al. [10].
- [50] Y. Ding, N. Robinson, S. Zhang, Q. Zeng, C. Guan, TSception: capturing temporal dynamics and spatial asymmetry from EEG for emotion recognition, *IEEE Transactions on Affective Computing* 14 (2023) 2238–2250. URL: <https://doi.org/10.1109/taffc.2022.3169001>. doi:10.1109/taffc.2022.3169001.
- [51] X. Ma, W. Chen, Z. Pei, J. Liu, B. Huang, J. Chen, A temporal dependency learning CNN with attention mechanism for MI-EEG decoding, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (2023).
- [52] M. P. Naeini, G. F. Cooper, M. Hauskrecht, Obtaining well calibrated probabilities using Bayesian binning, in: AAAI Conference on Artificial Intelligence, 2015. URL: <https://doi.org/10.1609/aaai.v29i1.9602>. doi:10.1609/aaai.v29i1.9602.
- [53] A. Salami, J. Andreu-Perez, H. Gillmeister, EEG-ITNet: an explainable inception temporal convolutional network for motor imagery classification, *IEEE Access* 10 (2022) 36672–36685. URL: <https://doi.org/10.1109/access.2022.3161489>. doi:10.1109/access.2022.3161489.
- [54] L. A. W. Gemein, R. T. Schirrmester, P. Chrabąszcz, D. Wilson, J. Boedeker, A. Schulze-Bonhage, et al., Machine-learning-based diagnostics of EEG pathology, *NeuroImage* 220 (2020) 117021. URL: <https://doi.org/10.1016/j.neuroimage.2020.117021>. doi:10.1016/j.neuroimage.2020.117021.
- [55] P. Ablin, J.-F. Cardoso, A. Gramfort, Faster independent component analysis by preconditioning with Hessian approximations, *IEEE Transactions on Signal Processing* 66 (2018) 4040–4049. URL: <https://doi.org/10.1109/tsp.2018.2844203>. doi:10.1109/tsp.2018.2844203.
- [56] L. Pion-Tonachini, K. Kreutz-Delgado, S. Makeig, ICLabel: An automated electroencephalographic independent component classifier, dataset, and website, *NeuroImage* 198 (2019) 181–197. URL: <https://doi.org/10.1016/j.neuroimage.2019.05.026>. doi:10.1016/j.neuroimage.2019.05.026.

doi:10.1016/j.neuroimage.2019.05.026.

- [57] T. Castermans, M. Duvinage, G. Cheron, T. Dutoit, About the cortical origin of the low-delta and high-gamma rhythms observed in EEG signals during treadmill walking, *Neuroscience Letters* 561 (2014) 166–170. URL: <https://doi.org/10.1016/j.neulet.2013.12.059>. doi:10.1016/j.neulet.2013.12.059.
- [58] J. E. Kline, H. J. Huang, K. L. Snyder, D. P. Ferris, Isolating gait-related movement artifacts in electroencephalography during human walking, *Journal of Neural Engineering* 12 (2015) 046022. URL: <https://doi.org/10.1088/1741-2560/12/4/046022>. doi:10.1088/1741-2560/12/4/046022.
- [59] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, V. Lempitsky, Domain-adversarial training of neural networks, *Journal of Machine Learning Research* 17 (2016) 1–35. URL: <https://www.jmlr.org/papers/v17/15-239.html>.
- [60] M. Long, Y. Cao, J. Wang, M. I. Jordan, Learning transferable features with deep adaptation networks, in: *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 2015. URL: <https://arxiv.org/abs/1502.02791>.
- [61] B. Sun, K. Saenko, Deep CORAL: correlation alignment for deep domain adaptation, in: *ECCV Workshops*, 2016. URL: https://doi.org/10.1007/978-3-319-49409-8_35. doi:10.1007/978-3-319-49409-8_35.
- [62] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, J. W. Vaughan, A theory of learning from different domains, *Machine Learning* 79 (2010) 151–175. URL: <https://doi.org/10.1007/s10994-009-5152-4>. doi:10.1007/s10994-009-5152-4.

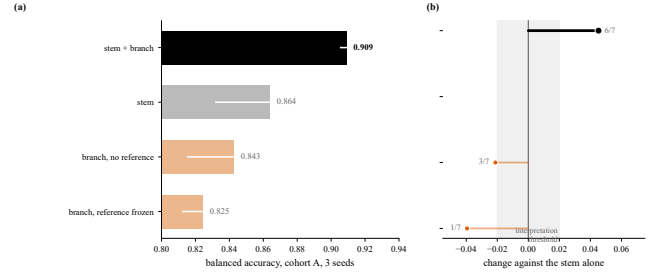


Figure 4: The branch against the stem on cohort A, three seeds. (a) Balanced accuracy, with the proposed model solid and the stem grey; white bars are one standard deviation across seeds. (b) The same as a paired change against the stem, with the annotation giving how many participants improved and the shaded band marking ± 0.02 , the descriptive run-to-run variation.

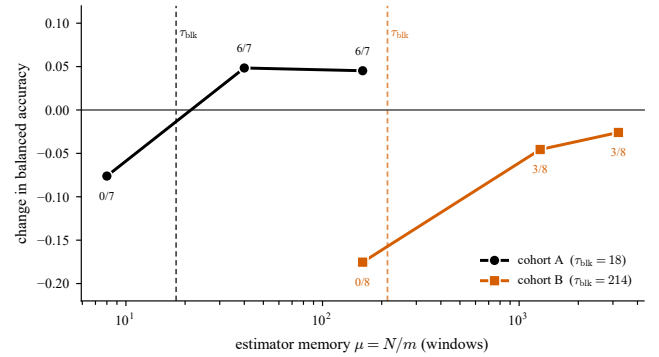


Figure 5: The rate condition measured by manipulation. Each point moves only the estimator memory $\mu = N/m$, with the architecture, the data and every other setting fixed; dashed verticals mark each cohort’s class-block length and annotations give how many participants improved. The branch gains at memories above the block length and reverses below it.

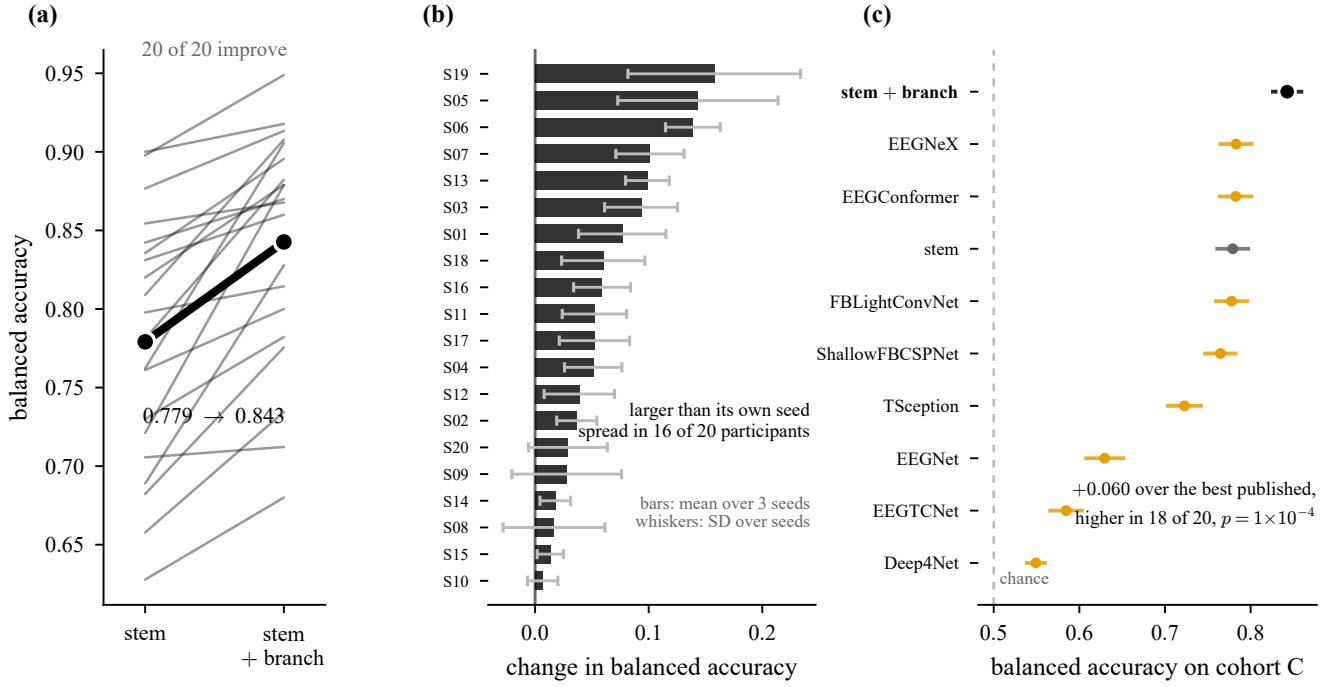


Figure 6: The branch on cohort C, held out from architecture and pipeline development, three seeds throughout. (a) Balanced accuracy, one line per participant, with the mean heavy. (b) The same change per participant, sorted; whiskers are that participant's standard deviation across seeds. (c) The eight published decoders of Table 3 on the same cohort, as mean and standard error over participants.

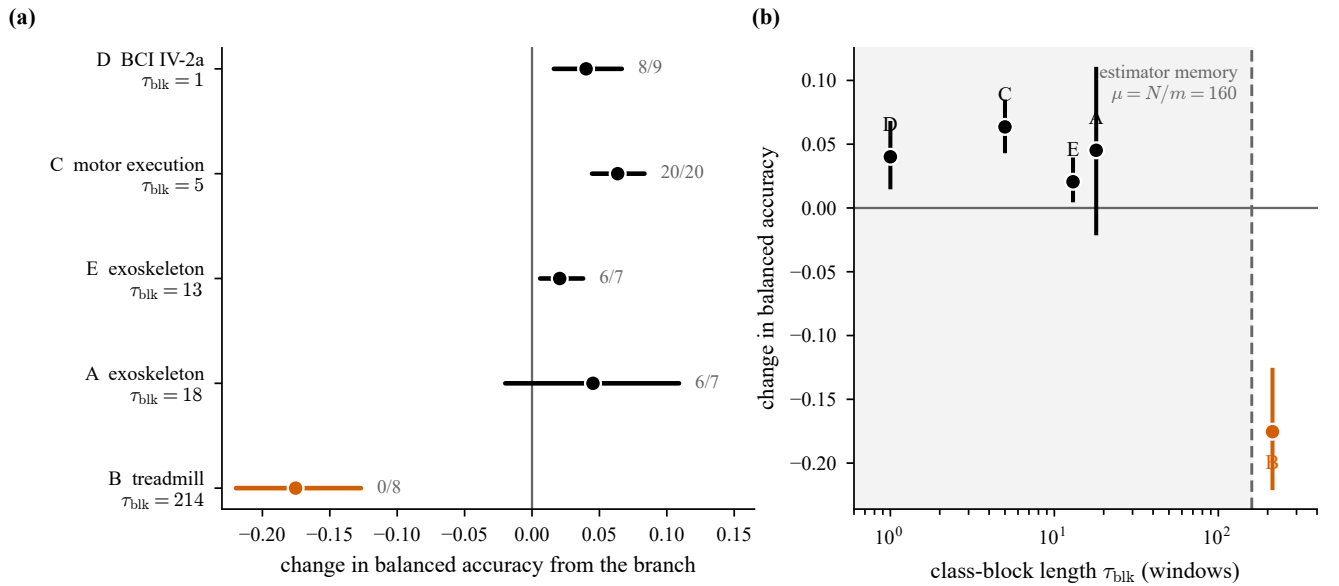


Figure 7: The branch on every cohort. (a) Change in balanced accuracy against the stem alone, paired across participants; bars are 95% paired-bootstrap intervals and the annotation gives how many participants improved. Cohorts are ordered by class-block length. (b) The same five effects against that block length, with the estimator memory dashed.